NSE/

ENGINE BLUEPRINT

Everything Cursor needs to build the NSE equity engine as the crypto engine's twin: same portal, same paper trading, same evidence discipline. Only the market layer changes. Review, market research, the opening and sentiment tracker, data model, API, design system, every page, phases, tickets and prompts.



OWNER · Dhruv Parekh (dhruvparekhdp)
PLANNER · Claude · 23 Oct 2026
BUILDER · Cursor
REPO · NSE-TRADE (private) · branch feat/mvp-backtest-news
SOURCE OF TRUTH · docs/PLAN_V2.md + docs/MARKET_INDIA.md + docs/PLAN.md

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A Appendix A · Full build plan (PLAN_V2.md): data model, settings, API, phases, tickets, Cursor prompts, owner questions, risks

B Appendix B · Full market research (MARKET_INDIA.md) with sources

How to read this

Sections 1–19 are the designed overview: diagrams, the design system and every page with numbered callouts. Appendices A and B are the complete build documents, printed from the same files Cursor reads. If anything here disagrees with `docs/PLAN_V2.md`, the file wins and this PDF gets regenerated.

1 · Read me first

OWNER

Decides. Answers the questions in Appendix A §11. Approves every phase PR. Gives the written yes for anything live. Runs things that need his secrets or machine (bhavcopy download, Angel app checks).

CLAUDE (PLANNER)

Wrote this plan, reviews every Cursor PR against it, keeps `docs/DECISIONS.md` current. Does not write the product code.

CURSOR (BUILDER)

Builds one phase per branch from the prompt in Appendix A §10. Test first for every ticket. Asks instead of guessing. Ends each phase with `docs/phase-reports/V<N>.md`.

Rules that never bend (AGENTS.md)

- Real orders need `mode=live` **and** `live_orders=true`, set by the owner in the server .env. Code and tests never set them.
- No secrets in code, tests, logs, commits or chat. Keys live encrypted in Postgres (/keys) or the server .env.
- Strategies, costs, risk and exits are pure functions shared by the lab and the engine. Paper must equal the backtest on the same bars.
- Research only through `lab/evaluate.py`, counted in the ledger, pre-registered in `docs/PREREG.md`. A change after seeing results is a new, post-hoc trial.
- UTC epoch ms stored, IST shown. Prices on the tick. LIMIT / STOPLOSS_LIMIT only (Angel bans market and IOC for algos). 5 orders per second.
- No deploy Mon–Fri 09:00–15:45 IST. Under 600 MB RAM on the server.

Order of work

PHASE	BRANCH	WHAT	WHY FIRST / DEPENDS ON
V0	feat/v0-hygiene	commit, safety, tick rounding, one data path, Test Angel button	everything else builds on a clean, safe base
V1	feat/v1-market	sessions (CAS), holidays, square-off lead, costs, instruments, survivorship, UDiFF + corporate actions, events	market facts must be right before any number means anything
V1b	feat/v1b-opening	opening & sentiment recorder (owner's ask)	pre-open and breadth can't be downloaded later: start recording early
V2	feat/v2-lab-fixes	backtester correctness, one exit engine, real baselines, portfolio	today's backtests are wrong (B1–B6)
V3	feat/v3-research	real data, PREREG, 5 trials, verdict	decides what paper trades
V4	feat/v4-paper	Book B end-of-day paper engine (crypto paper cycle)	needs V1–V3
V5	feat/v5-portal	the 7-page portal (crypto micro-UI)	can start after V0, against fixtures
V6	feat/v6-intraday-shadow	Angel WebSocket, 1m recording, Book A shadow	after V4
V7	feat/v7-live	LIVE readiness (real orders behind both flags)	only after go-live gates + owner's written yes

2 · Summary

TODAY (23 OCT 2026)

- No paper trading runs: every trading job in the runner is a stub that only logs.
- All research data is synthetic (`source = "synth"`). The S6 report is a smoke test.
- Backtests are wrong in ways that flatter results: shorts never stopped (−30R in a probe), gap-down stops filled at the stop (−1.5R recorded vs about −10.6R real), mirror/random baselines invented.
- The paper broker never adds intraday profit or loss to cash.
- Four of the six intraday strategies can never fire: nothing supplies their inputs.
- Market rules moved: the closing auction (since 3 Aug 2026) ends F&O stocks' continuous trading at 15:15, and the 15:05 square-off may now be after Angel's. Muhurat 2026 is in the afternoon, not the evening.
- The portal has the crypto look but placeholder pages. Keys are typed over plain HTTP.
- Nothing is committed yet. Angel login fails (AB1050, TOTP/client mismatch).

TARGET

- The crypto engine's concept on NSE: one book paper trades (Book B daily swing, CNC), intraday (Book A) runs shadow only, every signal logged with mirror + random twins and replayed to an outcome, filters audited as taken vs skipped.
- A ₹ paper cycle wallet with costs split the Indian way: STT, stamp, exchange, SEBI, GST, DP, brokerage.
- A market layer that knows sessions by date and stock class, holidays, ticks, bands, T+1, surveillance and corporate actions.
- An opening and sentiment recorder: GIFT Nifty, US, Europe, Asia, FX, crude, FII/DII, India VIX, tagged war and macro news, the pre-open auction and the first 30 minutes. Information first, a filter only if the lab proves it.
- The crypto "Brutal Command" portal, **clubbed to 7 pages**.
- Go-live only through gates, and only with the owner's written yes.

Decisions proposed (owner confirms; Appendix A §2)

#	DECISION	WHY
D1	Book B paper trades first, on end-of-day data	paper equals backtest by construction; no live feed needed
D2	Book A shadow only	at ₹50k the ₹1L minimum needs ≤ 0.5% stops, which cost 0.29R; SEBI: 71% of intraday traders lose
D3	Paper capital ₹1,00,000, min CNC position ₹20,000	₹10k positions cost 0.79% a round trip (DP + brokerage floor)
D4	Square off ≥ 15 min before the broker's cut-off	CAS moved cut-offs to about 15:00–15:10
D5	Filters and new strategies only as pre-registered trials	crypto lesson: filters rarely help; count every try
D6	One fill/exit engine for lab, paper and outcomes	no drift between backtest and paper
D7	Port crypto's portal micro-UI	owner wants the same UI
D8	Record the opening and sentiment from early on, information only	owner: "opening is a very big thing"; crypto: news echoed price

3 · Crypto → NSE: what stays, what changes

The crypto engine reached its current shape through a lot of evidence: one swing book that is allowed to paper trade, intraday quarantined to shadow, every signal logged with mirror and random baselines, every signal replayed to an outcome (traded or not), filters audited as "taken vs skipped", a wallet-based paper cycle, a go-live checklist, and a four-hub "Brutal Command" portal. The NSE engine copies all of that. Only the market layer is different.

CONCEPT	CRYPTO ENGINE (REFERENCE)	NSE ENGINE (TARGET)
The book that paper trades	Swing book: 4h/8h breakouts, stop 3×ATR, target 3R, 7-day limit (`analysis/swing_book.py`)	**Book B**: daily breakouts S6/S6b (PLAN A10), CNC, long only, stop 2.5×ATR14 trailed, 20-session limit
Quarantined family	15m detectors run shadow only	**Book A** (S1–S5 intraday MIS) runs **shadow only**: signals and virtual outcomes, no paper trades, until it passes the gate (§2 D2)
Market filter	Skip swing signals when BTC 30-day vol is in the top third of its past year (`analysis/regime_gate.py`)	Same idea on **Nifty 50** 30-day realised vol, as a pre-registered **trial** first (V3), not switched on by assumption
Paper wallet	Paper cycle: starting wallet, target wallet, peak, equity, unrealised, hit rate, net P&L, costs (`analysis/paper_cycle.py` , Book › Paper cycle)	Same cycle object in ₹. Costs split into brokerage, STT, exchange, SEBI, stamp, GST, DP, slippage
Fill rule shared by backtest and paper	`resolve_candle` / `virtual_outcome`	New `core/exits.py` , used by the backtest, the paper engine and the outcome replay (V2-7)
Every signal resolved	`_swing_outcomes_job` replays traded and skipped signals	`job_outcomes` replays every pick, mirror, random and skipped signal on daily bars
Costs	Fees + 18% GST + slippage + funding	STT, stamp, exchange, SEBI, GST, DP, brokerage with min ₹5, slippage (`MARKET_INDIA.md` §5)
Sessions	24/7	Pre-open, continuous, closing auction, holidays, Muhurat. Times are dated and per stock class (`MARKET_INDIA.md` §2)
Leverage	Up to 10x	None for CNC. MIS ≤5x (shadow only for now)
Shorting	Either side	CNC long only. MIS either side
Settlement	Instant	T+1. Book B holds at least one night
Data	Binance klines + WS	NSE bhavcopy (UDiFF) + Angel SmartAPI candles, quotes and WS. Yahoo only for the ticker tape
Portal	4 hubs: Command / Book / Evidence / System; micro-UI `core.js` ; 4 palettes	Same hubs, pages, components and palettes; NSE data
Go-live	System › Go-live gates page	Same page, with NSE gates (§8)
Alerts	Telegram	Telegram, separate bot (unchanged)

4 · Review: what works, what's broken

3.1 WHAT WORKS

- Settings, the dual live flags and the pytest guard. Live order methods are hard-gated (`broker/angel.py:107-110`).
- Cost model formulas (`core/costs.py`) match Angel's structure. Only the rates need refreshing (V1-4).
- Indicators (SMA, EMA, Wilder ATR and RSI, VWAP, Donchian, Keltner), SQLite migrations, ledger, keys page with an encrypted store, Telegram client, market-hours deploy guard.
- Tests: 54 pass. Ruff has 2 errors (unused imports in `scripts/_remote_check_keys_quotes.py:6` and `scripts/login_check.py:19`), so CI is red.

3.2 WHAT ISN'T THERE YET

- **No paper trading runs.** Every trading job in `engine/runner.py:34-59` only logs. `/api/today` returns zeros (`web/portal_api.py:29-33`). `/api/health` hard-codes `login_ok: false` (`web/app.py:38`).
- **All research data is synthetic.** Every file in `data/lake/bars_1d/` has `source == "synth"` . The S6 report (45 trades, −0.26R) is a smoke test, not evidence.
- **The portal is a skin with placeholders.** It has the crypto look and four hubs, but none of crypto's pages (swing positions in R, paper cycle tiles, signals with "why refused", taken vs skipped, go-live gates, session guard, accuracy, history), one palette instead of four, and no component kit.
- **Nothing is committed.** The whole tree is untracked on `feat/mvp-backtest-news` (only "Initial commit" exists).

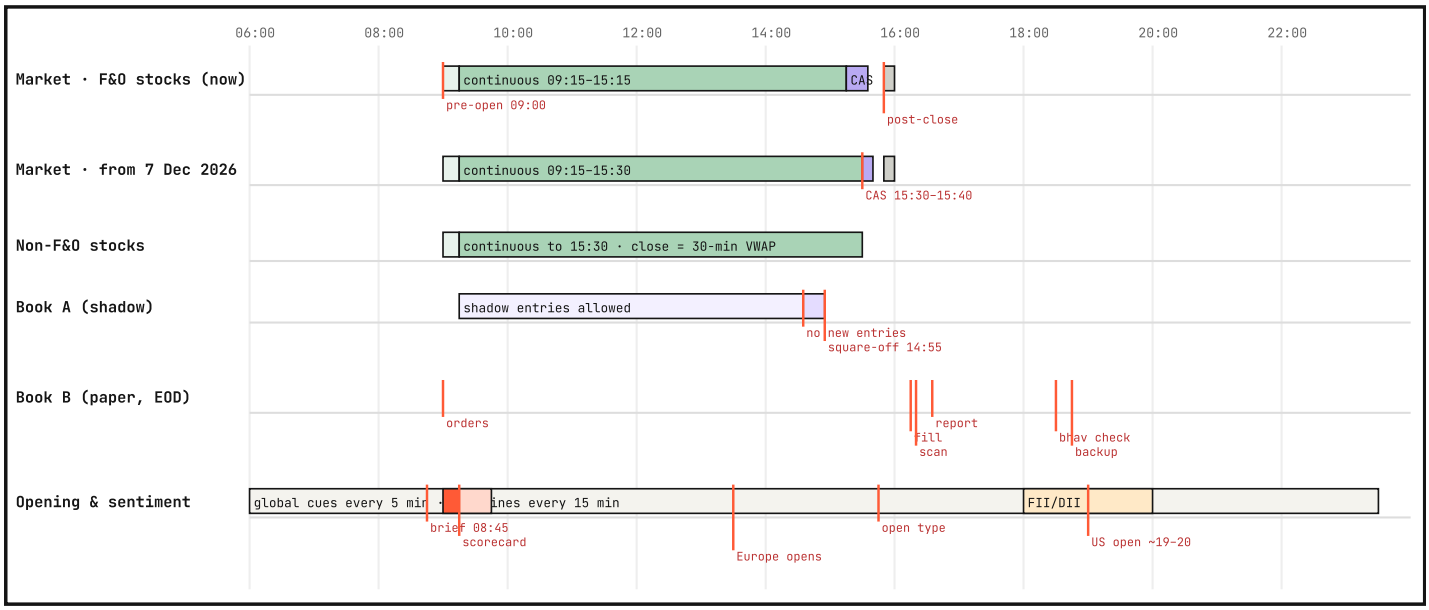
3.3 BUGS (CONFIRMED BY RUNNING PROBES ON 23 OCT 2026 UNLESS MARKED "BY READING")

#	SEVERITY	WHERE	WHAT IS WRONG	FIXED IN
B1	CRITICAL	`lab/backtest.py:189-197`	Intraday backtest checks the stop **only for longs**. A gap-fade short with its stop at 103.2 ran to 120.6: **−30R** recorded as one "square_off" trade	V2-2
B2	HIGH	`lab/backtest.py:157-230`	Intraday targets and `time_exit_ms` are ignored. Square-off happens at the last bar (15:29 in the probe), not 15:05. An ORB trade that reached 2R was held to the close	V2-2
B3	CRITICAL	`lab/backtest.py:78-83`	Daily stops always fill **at the stop price, even when the day opens far below it**. Probe: a −20% gap recorded −1.46R where the real fill (at the open) is about −10.6R. Backtests look far safer than reality	V2-1
B4	HIGH	`lab/backtest.py:114-124`	Daily entry assumes the next-day LIMIT (close +0.5%) always fills at that price. No check that the next day traded down to it, no better fill when it opens below, `entry_ms` set on the signal day. The trailing stop (PLAN A10) is missing	V2-1
B5	CRITICAL	`scripts/research.py:38-73`	**Mirror and random baselines are made up**: mirror R = −pick R, random R = 0. The gate's "beats mirror / beats random" checks reduce to "average R > 0". This breaks crypto lesson 2	V2-4
B6	HIGH	`scripts/research.py` , `lab/backtest.py:49`	Research runs one symbol at a time with its own capital. The n ≥ 100 per half gate can never be met, and there's no portfolio (shared cash, max open, priority)	V2-5
B7	CRITICAL	`broker/paper.py:154-161`	Paper MIS close returns only the margin. **Profit and loss never reach cash** (probe: +₹10,000 trade, cash unchanged at ₹1,00,000). Costs are never deducted. Partial reduces don't touch cash. Delivery buys can push cash negative	V4-3
B8	HIGH	`core/orders.py:239`	The stop watchdog measures the stop order's age, not how long price has been through the trigger. It forces an exit on the **first** crossing tick if the stop is more than 10 s old (probe confirmed). PLAN A9.4 says "crossed for longer than `stop_watchdog_s` "	V4-5
B9	HIGH	`core/orders.py:92,95,140,142,255`	Prices are rounded to 2 decimals, not the tick (probe: 1235.18 and 1216.11, invalid on a ₹0.10 tick). Breaks AGENTS rule 8. The exchange rejects such orders	V0-4
B10	MEDIUM	`core/orders.py:96-100`	"Idempotent by tag" can't work: a new tag is generated, then looked up	V4-5
B11	HIGH	`core/orders.py:114` vs `broker/angel.py:112,128`	Order manager calls `place_order(**kwargs)` / `cancel_order(client_tag)` . Angel takes `place_order(params: dict)` / `cancel_order(order_id, variety)` . `broker/base.py` is an empty async stub while both brokers are sync. "Paper uses exactly the live code path" isn't true yet	V4-5
B12	MEDIUM	`core/orders.py:268-276`	`square_off` cancels the stop and target but sends no exit	V4-5
B13	HIGH (BY READING)	`strategies/base.py:23-24`	`BarView.closed()` treats every bar as closed 60 s after its open. A daily bar stamped 09:15 counts as closed at 09:16, so a daily strategy run live would see the day's high, low and close hours early	V2-3
B14	HIGH (BY READING)	`strategies/orb.py:43-50` , `vwap_revert.py` , `intraday_momentum.py`	Strategies are fed 1m bars, but the rules are 5m/15m. ORB uses a 1m close and 1m volume against a 5m slot average. ATR is on 1m bars where PLAN says 5m	V2-2
B15	CRITICAL (BY READING)	`strategies/gap.py` , `pdh_rs.py` , `intraday_momentum.py` , `orb.py`	**Nothing ever supplies `prev_close` , `pdh` , `rs_rank` , `ret_decile` , `slot_vol_avg` or `gap_pct` ** (no caller in `lab/` , `engine/` , `scripts/` or `tests/`). S3, S3b, S4 and S5 can never fire. S1's volume filter is silently skipped, so it isn't the pre-registered rule	V2-2
B16	HIGH	`core/clock.py:55-102`	One fixed 15:30 close for every stock. Since 3 Aug 2026, F&O stocks stop continuous trading at 15:15 (CAS), and from 7 Dec 2026 at 15:30 with CAS to 15:40. No pre-open phases	V1-1
B17	HIGH	`config/holidays_2026.json:23-32`	Muhurat placeholder 18:15–19:15. The real 2026 session is in the **afternoon** (14:20 start)	V1-2
B18	HIGH	`data/calendar.py:77-80`	Only the 2026 file loads. From 1 Jan 2027 every weekday counts as a trading day. The runner's cron (`engine/runner.py:93-105`) runs Mon–Fri jobs on holidays too	V1-2
B19	HIGH	`settings.py:61-62` , PLAN A3	Square-off 15:05 can be later than Angel's post-CAS cut-off (reported 15:10 or earlier for F&O stocks)	V1-3
B20	MEDIUM	`config/costs_angel.json:10`	NSE transaction charge 0.00297%. Current tables show 0.00307%. Brokerage minimum ₹5 not modelled. `verified_against_calculator: false`	V1-4
B21	HIGH	`data/history.py:23-50`	Bhavcopy parser reads only the legacy columns. NSE's file has been UDiFF since July 2024. No split/bonus/dividend adjustment anywhere, so a 1:1 bonus looks like a −50% gap	V1-7
B22	MEDIUM	`data/free_quotes.py:22`	"Live" quotes are Yahoo daily candles (`interval=1d&range=5d`), delayed and unofficial, for 20 symbols. Fine for the tape, unusable for fills	V4-2
B23	MEDIUM	`scripts/research.py:25` , `engine/runner.py:26`	Research writes to `./data/engine.db` . The portal reads `./var/engine.db` (`settings.data_dir`), so research trials never show on the Lab page	V0-5
B24	HIGH	`config/settings.py:76` , DECISIONS 23 Oct	Portal binds `0.0.0.0` and is served publicly over **plain HTTP** (`http://52.62.37.4/nse/`). The admin password and Angel keys typed on `/keys` cross the internet unencrypted. Cookie has no `Secure` flag. PLAN A13 said Tailscale only	V0-2
B25	LOW	`core/orders.py:87,148` and similar	`now_ms or time.time()` treats 0 as missing (a test at ms=0 silently uses wall time)	V0-3
B26	LOW	`core/risk.py:104-107`	MIS exposure check ignores the new position's own value	V1-5
B27	LOW	`engine/runner.py:61-78`	News job ingests an empty list (never fetches). Fine while news is off. Leave it off	—

How the bugs were confirmed. Small hand-built inputs were run through the real code on 23 Oct 2026: a +₹10,000 MIS round trip left paper cash at ₹1,00,000; a gap-fade short with its stop at 103.2 exited at 120.6 (−30.2R); a −20% gap day filled a stop at the stop price (−1.46R); the watchdog fired on the first crossing tick; order prices came out as 1235.18 and 1216.11 (off a ₹0.10 tick). Cursor turns each of these into a failing test first (V2).

5 · The Indian market day, and the engine's day

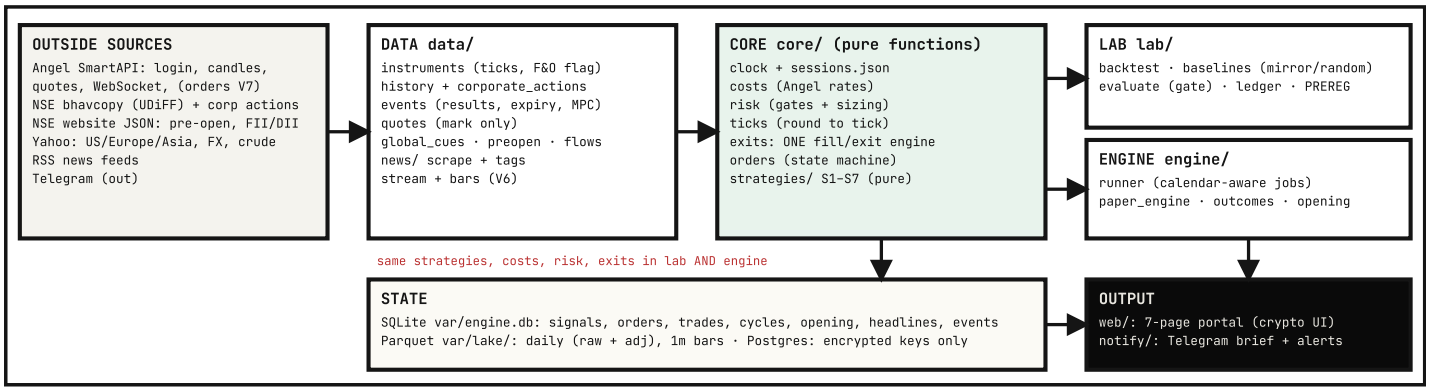
One picture of the session rules (MARKET_INDIA §2) and every engine job (Appendix A V1b and V4). All times IST.



RULE	VALUE	WHAT THE ENGINE DOES
Pre-open	09:00–09:15; order entry closes randomly 09:08–09:10 (since 7 Sep 2026)	records snapshots every 30 s; never trades it
Continuous, F&O stocks	09:15–15:15 until 6 Dec 2026; 09:15–15:30 from 7 Dec	dated regimes in sessions.json
Closing auction (CAS)	15:15–15:35 now; 15:30–15:40 from 7 Dec; close = auction price	never trades it; daily bar close flagged as auction
Non-F&O stocks	continuous to 15:30; close = 30-min VWAP	per-stock class
Post-close	15:50–16:00 at the closing price	not used
Broker MIS cut-off	Angel 15:15 (pre-CAS page); others 15:00–15:10 after CAS. VERIFY	engine square-off = broker − 15 min (14:55)
Muhurat 2026	Sun 8 Nov, afternoon (14:20 start; NSE circular). VERIFY	no new entries either book
Holidays	15 weekdays in 2026; 2027 list due around December	no year file = no trading + red health
Tick size	₹0.01 below ₹250 · 0.05 to ₹1k · 0.10 to ₹5k · 0.50 to ₹10k · 1 to ₹20k · 5 above	daily from the scrip master; round every price
Settlement	T+1; CNC long only; same-day CNC sell = intraday	Book B holds at least one night
Algo rules (1 Apr 2026)	static IP (change ≤ weekly), no market/IOC, 10 OPS threshold, daily 2FA	LIMIT / SL-L only, 5 OPS, IP check before first live order

6 · Architecture

Where data comes from, where the rules live, and what the owner sees. The core is pure functions used unchanged by the lab and the engine.



File-by-file changes

```
config/
  sessions.json            NEW  dated session regimes per stock class (V1-1)
  holidays_2026.json       FIX  Muhurat afternoon; schema adds "verified_by_owner" date (V1-2)
  holidays_2027.json       NEW  when NSE publishes (owner), loader requires it from 1 Jan 2027
  costs_angel.json         FIX  rates, min brokerage, source + date, verified_against_contract_note (V1-4)
  tick_bands.json          NEW  fallback tick table (MARKET_INDIA §4) (V0-4)
  universe_history.csv     NEW  Nifty 100 adds/removes with dates (V1-6)
  surveillance.json        NEW  ASM/GSM exclusions, refreshed daily (V1-5)
core/
  ticks.py                 NEW  round_to_tick(price, tick, direction) (V0-4)
  clock.py                 FIX  session_state(ms, *, stock_class) reads sessions.json (V1-1)
  exits.py                 NEW  the one fill/exit engine (V2-7)
  costs.py                 FIX  min brokerage; breakdown incl. slippage (V1-4)
  risk.py                  FIX  min_position_cnc_inr, exposure incl. new position, surveillance (V1-5)
  orders.py                FIX  tick rounding, real idempotency, watchdog timing, square-off exit (V4-5)
broker/
  base.py                  FIX  real sync Broker protocol + dataclasses (Order, Fill, Position, Funds, Quote) (V4-5)
  paper.py                 FIX  cash/P&L/costs accounting; CNC long-only; T+1 holding rule (V4-3)
  angel.py                 FIX  adapter to the Broker protocol (tag → order id map) (V4-5)
data/
  history.py               FIX  UDiFF + legacy bhavcopy, delivery file, corporate actions (V1-7)
  corporate_actions.py     NEW  adjustment factors (V1-7)
  events.py                NEW  results dates, expiry Tuesdays, RBI MPC, Budget (V1-8)
  quotes.py                NEW  Angel LTP poller for marking; Yahoo fallback for the tape (V4-2)
  global_cues.py           NEW  US / Europe / Asia indices + futures, USD/INR, DXY, Brent, gold, US 10y, GIFT Nifty (V1b-1)
  preopen.py               NEW  NSE pre-open snapshots (IEP, buy/sell qty) every 30 s, 09:00-09:15 (V1b-2)
  flows.py                 NEW  FII/DII provisional cash flows each evening (V1b-1)
news/
  india_rules.py           FIX  new event tags: geopolitics_india, middle_east, russia_ukraine, tariffs_trade, us_data,
                                china, opec, monsoon, gst_council, india_macro, block_deal, promoter_pledge (V1b-3)
  scrape.py                FIX  real fetch on a schedule, more feeds, published time from the feed (V1b-3)
engine/
  opening.py               NEW  morning brief, opening scorecard, opening type label, risk-off alarm (V1b-4)
lab/
  backtest.py              REWRITE on core/exits.py; portfolio mode across the universe (V2)
  baselines.py             NEW  real mirror + random trades (V2-4)
  evaluate.py              FIX  halves, by year, VIX tercile, results season, cost stress (V2-6)
  prereg.py + docs/PREREG.md NEW  trial definitions written before runs (V3-2)
engine/
  paper_engine.py          NEW  Book B EOD cycle: scan → prepare → fill → exits → mark (V4)
  outcomes.py              NEW  resolve every signal (taken/skipped/mirror/random) (V4-4)
  runner.py                FIX  real jobs, calendar-aware (V4-6)
  migrations/002_paper.sql NEW  §5
web/
  portal/ (core.js, portal.css, portal.html, nav.js, hubs/*.js)   PORT from crypto (V5)
  portal_api.py            FIX  the endpoints in §7
```

7 · UI design system (port of crypto "Brutal Command")

Brutalist structure on a black terminal ground. Thick off-white borders, hard offset shadows in the accent colour, square corners, mono data. Motion is decoration only and never carries a number. The real portal never shows simulated data. The rules come from `crypto/docs/PORTAL_REVAMP.md`.

Palettes (header swatches, saved per browser)

Sage & Lilac · default
acc #A9D3B6 · up #A9D3B6 · dn #B9AAF2

Champagne & Teal
acc #E3C58D · up #6CC9AE · dn #A99BF0

Ice & Peach · colour-blind safest
acc #94C5F5 · up #94C5F5 · dn #EDA982

Classic lime
acc #C6F432 · up #C6F432 · dn #9B8CFF

TOKEN	VALUE	USED FOR
--bg / --pn	#0A0A0A / #0F100E	page ground (40 px grid) / panels
--ink / --dim	#E8E6DF / #A3A39A	text, 3 px borders / labels, ages, notes
--acc	palette	logo block, tape, active tab, shadows, primary buttons, accents
--up / --dn	palette	profit, long, taken / loss, short, skipped, alarms
--on	#0A0A0A	text on accent fills
annotation	#FF5A36	this document's callouts only

Type and spacing

ROLE	FONT	SIZE (DESKTOP / PHONE)
Headings, big numbers	Archivo 900	tile value 26 / 22 px · panel title 13 / 12 px uppercase, letter-spacing .05em
Data, labels, tables, trace	JetBrains Mono 400/700	13 / 12 px · labels 10 / 10 px uppercase dim
Spacing	8 px grid	panel padding 12 px · gaps 16 px · borders 3 px (panels) / 2 px (tiles, chips)
Phone	≤ 480 px	rows stack; tap targets ≥ 40 px; tables scroll inside their panel, never the page

Component gallery (rendered with the real styles)



Plain-English reasons (`web/reasons.py`; the UI never shows a raw code)

CODE	SHOWN AS
kill_switch	kill switch is on: no new entries
floor	capital is below the floor (80% of start)
daily_loss_limit	today's loss reached the daily limit
session_closed	market is not open for new entries now
max_open_swing / max_open_intraday	the maximum number of positions is already open
mis_exposure	intraday exposure would pass 3× capital
symbol_open	already holding this stock
spread_wide	bid–ask spread is too wide right now
near_circuit	too close to today's price band

results_day	results due within 3 days
below_min_position	intraday position under ₹1,00,000: costs would eat too much
below_min_cnc_position	position under ₹20,000: costs would eat too much
surveillance	stock is on an NSE surveillance list (ASM/GSM)
market_filter	Nifty volatility in the top third of the past year
no_fill	the limit price wasn't reached the next day
calendar_missing	no holiday file for this year: engine won't trade
book_a_shadow	Book A is shadow only
reconcile_fault	our records and the broker's disagree for this stock

Motion

EFFECT	WHERE	RULE
glitching hub name, blinking cursor	logo	every ~3 s; decoration
scanline sweep	whole page	6 s loop; decoration
scrolling tape	top band	48 s loop, pauses on hover
pipeline hand-off	Today	1.6 s; never carries a number
trace types itself	Today	only real event lines
rows slide in, toast	feeds	once per new event
all motion	everywhere	off with prefers-reduced-motion

Accessibility

- Colour never carries meaning alone: LONG/SHORT words, ✓/✗, PASS/FAIL, signs on numbers.
- Text contrast ≥ 4.5:1 on the black ground; dim text only for secondary facts.
- Every stale value says its age ("stale 15 h"). Unknown is "—" or "no data", never 0.
- Keyboard: tabs and filters focusable; Esc closes drawers.

8 · Navigation: 21 pages clubbed into 7

Keep hash routing (`#/book/paper`) because the portal is served under `/nse/` . Everything else follows `crypto/docs/PORTAL_REVAMP.md` : the Store (poll only while a widget uses a key, pause in hidden tabs), widgets, views, the component kit `UI.C` , the four palettes (Sage & Lilac default, Champagne & Teal, Ice & Peach, Classic lime) with header swatches and `localStorage` , motion only as decoration, `prefers-reduced-motion` , and **nothing simulated on the real portal**.

Seven pages, not twenty-one (owner, 23 Oct 2026: "cut them short to only required, or club them"). The four crypto hubs stay, so the portal looks and navigates like crypto. Each hub holds only the pages that earn their place. Anything that was only another view of the same data is now a section or filter on one page.

HUB	PAGE (ROUTE)	WHAT IT CLUBS	SECTIONS, TOP TO BOTTOM
01 Command	**Today** (`#/command`)	Overview + Session guard + Live ₹ status	session hero (state, countdown to next change, square-off and no-entry times, kill switch); pipeline (data → strategies → risk → paper book → opening recorder); wallet mini-tiles (equity, today's P&L, open positions); guard flags (daily loss, floor, feed stale, reconcile, calendar, risk-off alarm); universe heat grid; decision trace; latest signal feed + toast
	Market mood (`#/command/mood`)	Opening + Sentiment + market context	morning brief (frozen 08:45); pre-open board (09:00–09:15 live); opening scorecard (from 09:15) and opening type (15:45); markets grid (GIFT Nifty, US, Europe, Asia, FX, crude, gold, bonds, India VIX, FII/DII); sentiment gauge with parts; risk-off banner; headlines timeline with tag chips; 60-day opening history
02 Book	**Paper book** (`#/book`)	Swing book + Paper cycle + Live ₹ (locked)	cycle tiles (equity, cash, unrealised, net P&L, hit rate, costs split, cycle #); equity curve; open positions as R cards (Book B; Book A appears only if promoted); tomorrow's pending orders; closed trades with book/strategy filters and totals; taken vs skipped; rules in force; a locked "Live ₹" strip showing which gates are still open
	Signals (`#/book/signals`)	Signals + Intraday shadow + History	filter bar (date range, book, strategy, role pick/mirror/random, decision taken/skipped/shadow, symbol search); summary tiles; one table for every signal incl. Book A shadow with virtual outcome; "why refused" side panel in plain words; CSV export (this replaces the History archive)
03 Evidence	**Evidence** (`#/evidence`)	Strategies + Accuracy + Research	gate table per trial (pass/fail chips, prereg link, trials counter); R histogram and win rate by strategy/month; paper-vs-backtest check; latest research reports with by-year, VIX tercile and results-season tables
04 System	**Health** (`#/system`)	Go-live + Health + Data + Diagnostics	go-live verdict and gates; jobs (last run, result, next run); data coverage (lake, bhavcopy, corporate actions, pre-open capture days, missing days); feed/source states with age and errors; memory vs 600 MB; recent events
	Settings (`#/system/settings`)	Settings + Keys + Watchlist + API list	runtime settings (admin, whitelisted, every change logged); derived session times (read-only); universe and manual exclude/include; keys (encrypted, set / not set) with the **Test Angel (read-only)** button; API list link

Retired as pages (still reachable as sections or filters): Overview, Opening, Sentiment, Watchlist, Swing book, Paper cycle, Intraday (shadow), Live ₹, Session guard, Strategies, Accuracy, History, Research, Go-live, Data, Diagnostics, Keys, API list. Old crypto-style addresses (`#/book/paper` , `#/command/signals` , `#/system/golive` …) redirect to the page and section that now hold them.

Copy `crypto/tests/test_portal.py` 's structural checks: every native page has a view, every view names existing widgets, every widget uses a defined Store key.

Page map

HUB	PAGE	ROUTE	POLLS (ONLY WHILE VISIBLE)
01 Command	Today	#/command	meta 5 s · paper 20 s · guard 10 s · quotes 15 s · signals 30 s
01 Command	Market mood	#/command/mood	opening 30–60 s · sentiment 60 s · headlines 5 min
02 Book	Paper book	#/book	paper 20 s · swing 30 s · golive 60 s
02 Book	Signals	#/book/signals	signals 30 s (paged)
03 Evidence	Evidence	#/evidence	lab 5 min · parity 5 min
04 System	Health	#/system	golive 60 s · health 30 s · data-health 5 min · events 60 s
04 System	Settings	#/system/settings	meta · keys (admin) on demand

9 · Header and chrome anatomy (on every page)



Callouts: 1 logo · 2 hub tabs · 3 IST clock · 4 session chip · 5 mode chip · 6 palette swatches · 7 status strip · 8 tape · 9 page tabs · 10 toast

1. **Logo**: `NSE/` block in the accent colour + the hub name with the blinking cursor and a glitch every ~3 s (decoration only). Click → `#/command`.
2. **Hub tabs**: 01 COMMAND · 02 BOOK · 03 EVIDENCE · 04 SYSTEM, from `web/portal/nav.js` (the one nav map). Active hub is filled with the accent and has a hard 2 px shadow.
3. **IST clock**: Browser clock converted to Asia/Kolkata, ticks every second. Stored times are UTC ms; every time on the portal is IST.
4. **Session chip**: From `/api/portal/meta.session`: PRE-OPEN (shows "random close 09:08–09:10"), OPEN, NO NEW ENTRIES, SQUARE-OFF, CLOSING AUCTION, POST-CLOSE, CLOSED, HOLIDAY (with its name), SPECIAL (Muhurat). Second line: countdown to the next change. It uses the F&O-stock regime, because most of the universe is F&O.
5. **Mode chip**: PAPER (ink fill) or LIVE ₹ (red fill + the word REAL). Never changes from the browser.
6. **Palette swatches**: Sage & Lilac (default), Champagne & Teal, Ice & Peach, Classic lime. Sets `data-palette` on <html> and saves `nse-palette` in localStorage. An inline head script applies it before first paint.
7. **Status strip**: Short facts that must always be visible: calendar loaded and verified for this year (red if not), data freshness (last bhavcopy date, quote age), Angel login state, kill switch, market filter, risk-off alarm.
8. **Tape**: Scrolling Nifty 50, India VIX and the universe's LTP and % change from `/api/quotes` (Angel LTP; Yahoo fallback). Stale quotes (> 2× poll interval) turn grey with their age. Pauses on hover.
9. **Page tabs**: The current hub's pages (1–2 per hub). Hash routes, so the portal works under `/nse/`.
10. **Toast**: Full-width band for new events: SIGNAL LOCKED · ORDER PREPARED · FILLED · STOPPED · SKIPPED (reason) · RISK-OFF. Slides in, 3 s, one at a time, never stacks.

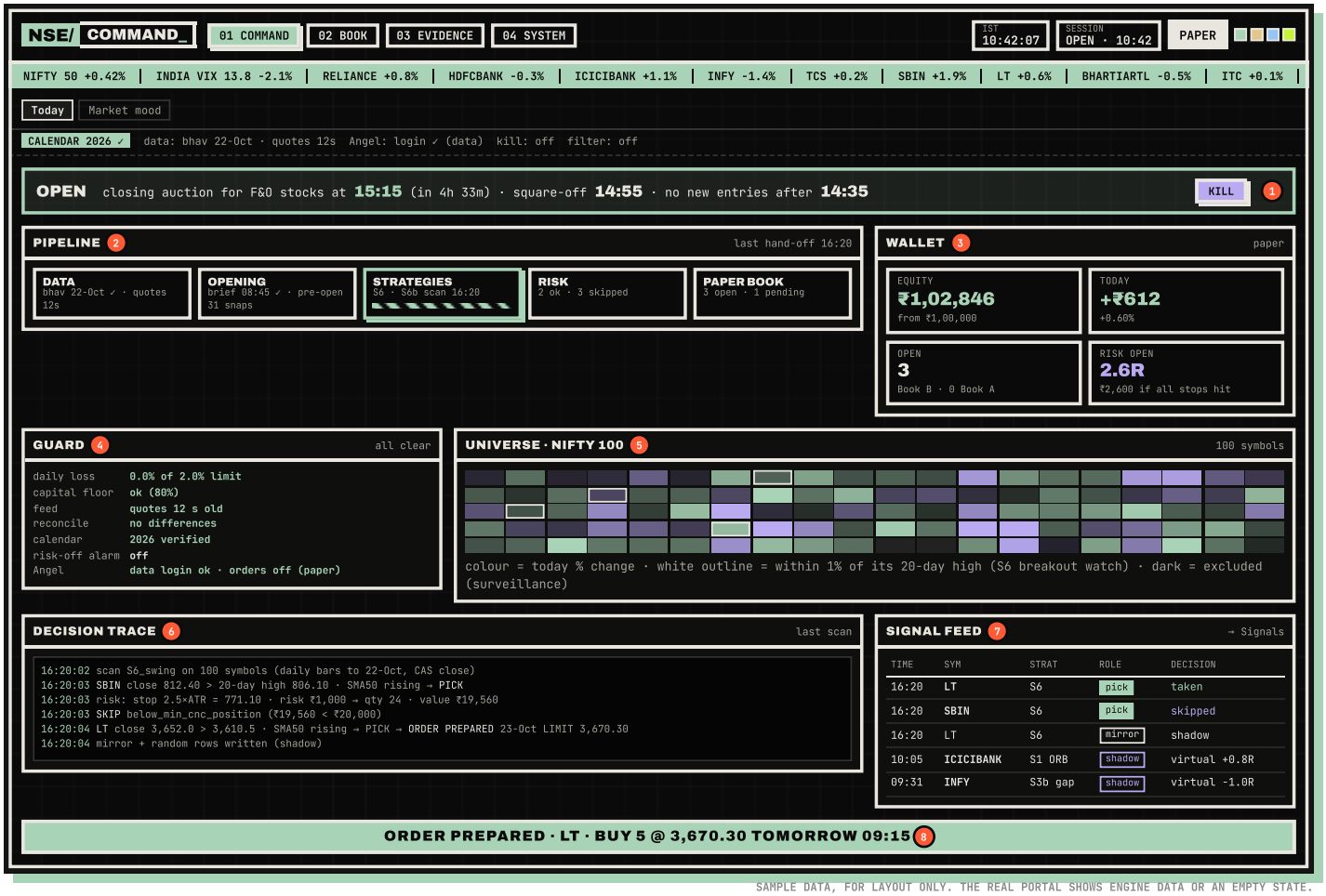
10 · Today: what is happening right now

`#/command · Command › Today`

CLUBS — Overview + Session guard + Live ₹ status (crypto: Command › Overview, Book › Session guard)

FOR — The first screen every time; the phone screen during market hours.

DATA — `/api/portal/meta` (5 s) · `/api/paper` (20 s) · `/api/guard` (10 s) · `/api/quotes` (15 s) · `/api/portal/signals?days=2` (30 s) · `/api/events` (60 s)



SAMPLE DATA, FOR LAYOUT ONLY. THE REAL PORTAL SHOWS ENGINE DATA OR AN EMPTY STATE.

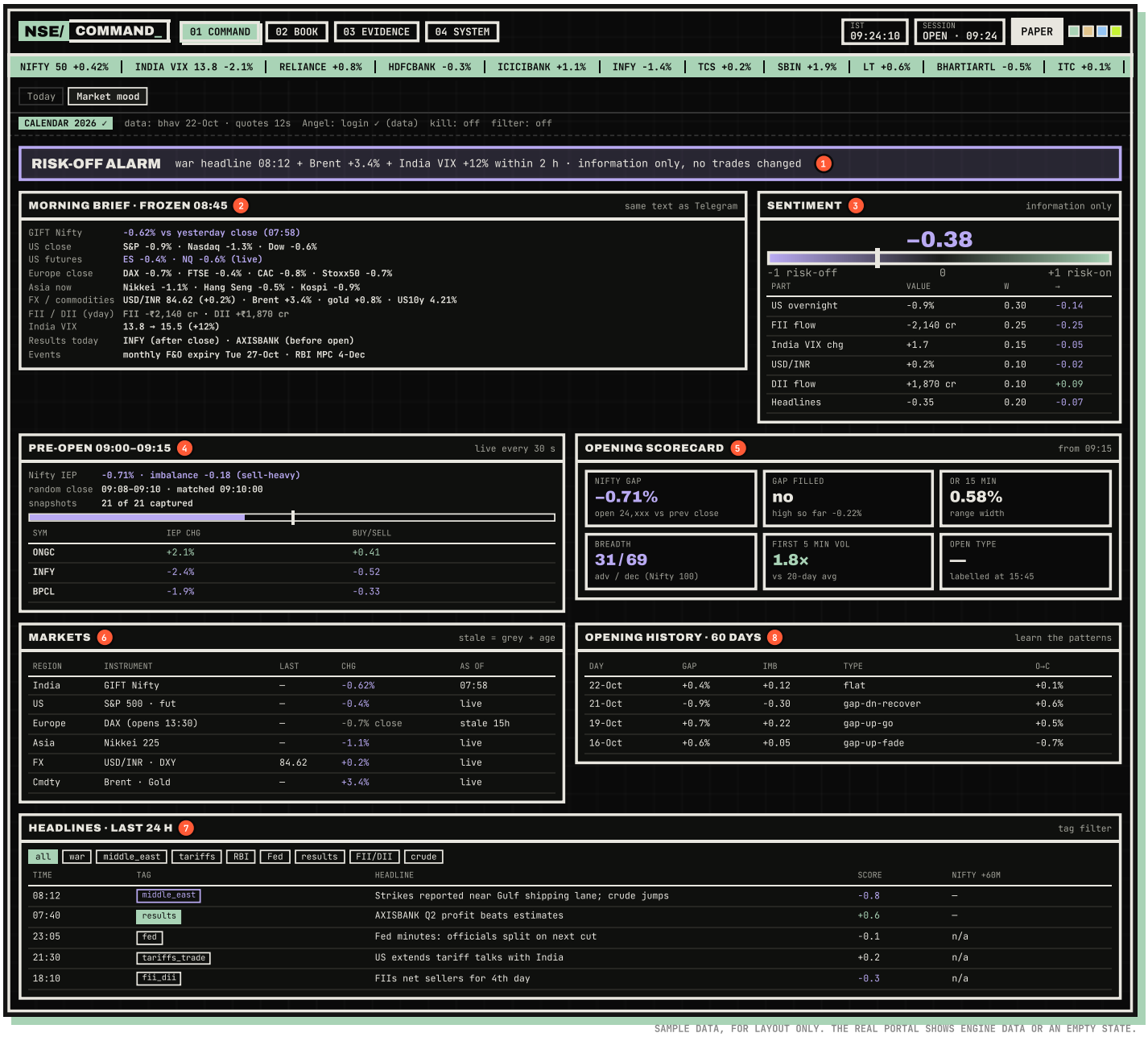
1. **Session hero**: State in big type, the next session change with a countdown (from `sessions.json` for today's regime), derived square-off and no-entry times, and the **KILL** button (admin cookie, asks "Stop all new entries?", writes an event). With the kill switch on, the band turns lilac/red: "KILL SWITCH ON: no new entries, open positions keep their stops" + RESUME.
2. **Pipeline**: Five agents with their real last-run facts: DATA (last bhavcopy, quote age), OPENING (brief built, pre-open snapshots), STRATEGIES (last scan, strategies enabled), RISK (ok/skipped counts), PAPER BOOK (open, pending). The working card's highlight moves every 1.6 s as decoration only. A job that failed shows its card in the down colour with the error in one line.
3. **Wallet**: Equity (cash + marked holdings), today's P&L after costs, open count by book, and open risk (sum of the distances to stop × qty, also in R). Tap → Paper book.
4. **Guard**: Every safety rail in one list (PLAN A12.3): daily loss vs limit, capital floor, feed age, reconcile differences, calendar status, risk-off alarm, Angel state. Any red line also lights the status strip.
5. **Universe heat grid**: 100 squares = today's Nifty 100 (from `universe_on(today)`). Colour = % change; outline = within 1% of its 20-day high (where S6 fires); dark = excluded or under surveillance. Tap a square → that symbol's signals.
6. **Decision trace**: The last scan's log in plain words, typed out like a terminal: rule checked, values, risk maths, decision and reason. Comes from the `events` table (kind `decision`), never invented.
7. **Signal feed**: Newest signals first (pick, mirror, random, shadow) with their decision; new rows slide in. Header link → Signals.
8. **Toast**: Shown once per new event (see the header anatomy).

Empty states. Before the first scan: "No signals yet. The first scan runs at 16:20 on a trading day." No paper cycle: "Paper book is off. Turn it on in System › Settings." Holiday: the hero says "HOLIDAY · Diwali-Balipratipada · next session Wed 11 Nov 09:00".

11 · Market mood: the opening and the market's sentiment

`#/command/mood · Command › Market mood`

CLUBS	Opening + Sentiment + market context (new in V1b, owner 23 Oct: "track entire sentiment with GIFT Nifty, US and Europe market, war news, every major thing")
FOR	08:30–09:45 every trading day; later to look back at how the day was set up.
DATA	`/api/opening` (30 s in pre-open, 60 s otherwise) · `/api/sentiment` (60 s) · `/api/headlines?hours=24` (5 min)
RULE	`INFORMATION ONLY.` Nothing on this page blocks, sizes or times a trade until a pre-registered lab trial shows lift.



SAMPLE DATA, FOR LAYOUT ONLY. THE REAL PORTAL SHOWS ENGINE DATA OR AN EMPTY STATE.

1. **Risk-off banner**: Appears only when the alarm rule fires (a war/geopolitics headline **and** Brent ≥ +3% **and** India VIX ≥ +10% within 2 h; thresholds in Settings). One Telegram alert per day. Says plainly that no trade was changed.
2. **Morning brief**: Built and **frozen at 08:45** (`morning_brief` table) so it can be checked against what happened. GIFT Nifty (or "proxy: US futures + Asia" until a source is set), US close and futures, Europe close, Asia live, USD/INR, DXY, Brent, gold, US 10y, FII/DII, India VIX, results and events today. Identical text goes to Telegram.
3. **Sentiment gauge**: Composite −1…+1 from `news/sentiment.composite_score` with every part, its weight and its contribution visible. Missing parts are listed as "no data", never as 0.
4. **Pre-open board**: Live 09:00–09:15: Nifty indicative price vs previous close, buy/sell imbalance bar (−1 sell-heavy … +1 buy-heavy), the random-close window, snapshots captured vs expected, and the top pre-open movers. After 09:15 it freezes on the final match.
5. **Opening scorecard**: From 09:15: Nifty gap, gap filled (and when), opening range width, breadth (advancers/decliners in the Nifty 100), first-5-minute volume vs its 20-day average, and the opening type label written at 15:45 (gap-up-go / gap-up-fade / gap-down-recover / gap-down-go / flat).
6. **Markets grid**: Every instrument in `config/cues.json` grouped by region with last, % change and age. A closed market shows its last close in grey with the age ("stale 15 h"), never a fake 0%.
7. **Headlines timeline**: Tagged headlines (war, middle_east, russia_ukraine, tariffs_trade, RBI, Fed, us_data, results, FII/DII, crude, monsoon…) with source link, score and the Nifty move in the 60 minutes after it (filled in later). Tag chips filter. Duplicates are merged.
8. **Opening history**: The last 60 trading days: gap, final imbalance, opening type and open-to-close, so the owner can see patterns. It becomes a research dataset (V1b-6).

12 · Paper book: the wallet, the positions, the result

`#/book · Book › Paper book`

CLUBS	Swing book + Paper cycle + Live ₹ (crypto: Book › Swing book, Book › Paper cycle, Book › Live ₹)
FOR	Evenings after 16:35 (fills and exits are end-of-day), weekends for review.
DATA	`/api/paper` (20 s) · `/api/swing` (30 s) · `/api/golive` (60 s)
RULE	Fills and exits come only from `core/exits.py` on daily bars, the same code as the backtest. Intraday quotes only mark positions for display.



SAMPLE DATA, FOR LAYOUT ONLY. THE REAL PORTAL SHOWS ENGINE DATA OR AN EMPTY STATE.

1. **Cycle tiles**: Equity = cash + holdings × mark. Cash = what is free for new buys. Unrealised, net P&L after all costs, hit rate with average R, costs paid (split shown in 3), **cost per trade in R** (the number that killed crypto intraday; it goes amber above 0.15R), cycle number and age. Crypto's target-wallet tile appears only if `paper_target_wallet` is set.
2. **Equity curve**: One point per trading day after costs, with the starting line. Hover/tap shows date, equity and the day's closed trades.
3. **Costs paid**: Bars per charge since the cycle began: STT, brokerage, DP, slippage, GST, stamp + exchange + SEBI. Uses the exact `costs_json` stored on each trade.
4. **Open positions**: One card per position: symbol, side (always LONG for CNC), strategy, **R now** in big type, the stop → entry → trail bar, initial stop vs current trailed stop, entry and mark, sessions left of the 20-session limit, unrealised ₹. Book A cards appear only after Book A is promoted to paper.
5. **Tomorrow's orders**: `pending_orders` written by the 16:20 scan: LIMIT price (on the tick), stop, qty, status. Admin can cancel one (logged).
6. **Live ₹ (locked)**: A one-line strip, not a page: how many go-live gates are open, with a link. When live is allowed in V7 this strip becomes the live book's summary.
7. **Closed trades**: Sortable; book and strategy chips filter; header shows count, win rate, average and total R. The exit reason is in words ("stop (gap open)", "trail stop", "20 sessions"). Tap a row → the trade's signal, fills and cost split.
8. **Taken vs skipped**: Crypto's filter audit: every taken and every skipped signal resolved by the same exit engine, so you can see whether the skip rules (min size, max open, results day…) help.
9. **Rules in force**: The exact rules this cycle runs (from the cycle's `config_json`), so a change starts a new cycle instead of silently mixing results.

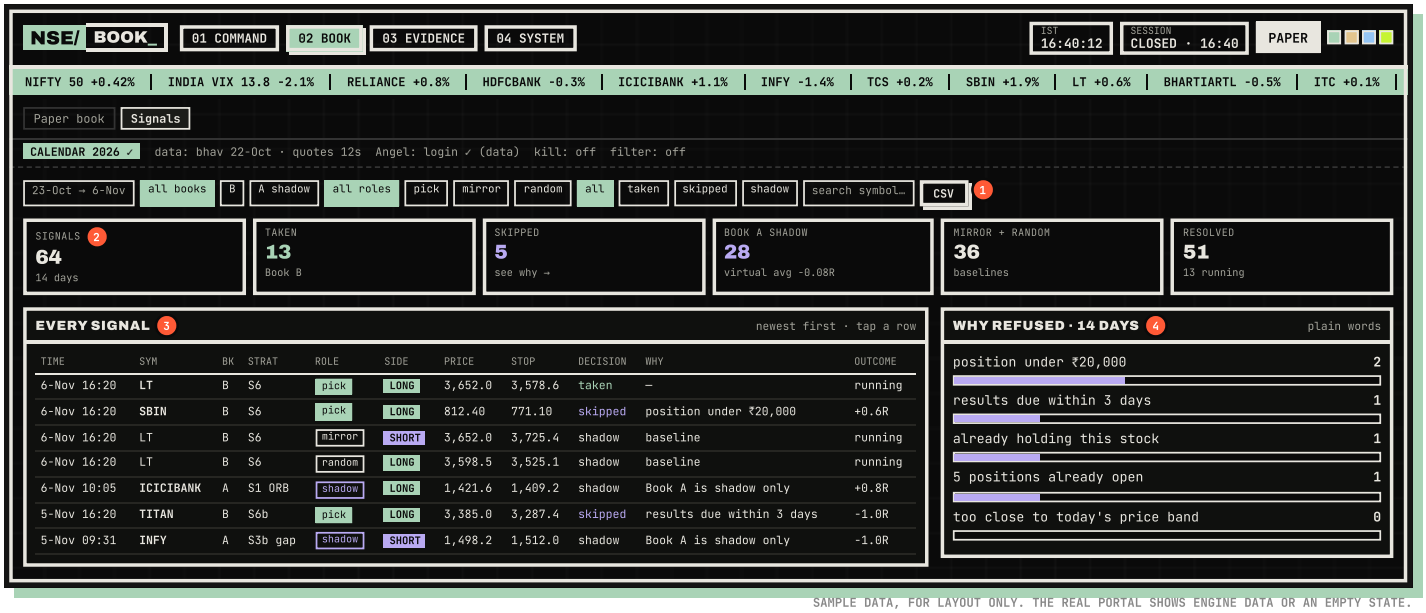
13 · Signals: every decision, taken or not

`#/book/signals · Book › Signals`

CLUBS — Signals + Intraday (Book A shadow) + History archive (crypto: Command › Signals, Evidence › History)

FOR — Checking why something was or wasn't traded; auditing Book A shadow.

DATA — `/api/portal/signals?from=&to=&book=&strategy=&role=&decision=&q=` (30 s, paged 200 rows) · CSV = same query with `&format=csv`



SAMPLE DATA, FOR LAYOUT ONLY. THE REAL PORTAL SHOWS ENGINE DATA OR AN EMPTY STATE.

1. **Filter bar**: Date range (default 14 days; any range replaces the old History page), book, role (pick/mirror/random/shadow), decision (taken/skipped/shadow), symbol search, CSV export of exactly what is shown. Filters live in the URL so a view can be bookmarked.
2. **Summary tiles**: Counts for the current filter: signals, taken, skipped, Book A shadow (with its virtual average R), mirror + random baselines, resolved vs running.
3. **Every signal**: One row per signal row in the DB. Price/stop on the tick. DECISION and WHY come from `skip_reason` through `web/reasons.py` (never raw codes). OUTCOME = the virtual R from the outcome job for every row, so skipped and shadow signals are judged the same way as trades. Tap a row → detail drawer: features at decision time (gap, ATR %, India VIX, Nifty vol rank, results-in-days, sentiment composite, opening type so far), the bars around it, and its mirror/random twins.
4. **Why refused**: Count of each skip reason in plain words for the filter period, sorted by count. A reason that blocks most signals is a design question for the owner, not a bug to silence.

14 · Evidence: what the lab and the paper book have proven

`#/evidence · Evidence`

CLUBS Strategies + Accuracy + Research (crypto: Evidence › Strategies, Accuracy, Research)

FOR Deciding what may paper trade or go live. The owner's gate decisions are made here.

DATA `/api/lab` (5 min) · `/api/paper` · `/api/parity` · reports from `var/reports/*.json`



SAMPLE DATA, FOR LAYOUT ONLY. THE REAL PORTAL SHOWS ENGINE DATA OR AN EMPTY STATE.

1. **Counters**: Trials so far (every lab run, from the ledger; this number never goes down), how many passed, and how much data each book has. Book A shows 0 days until the live recorder has history.
2. **Gate table**: One row per pre-registered trial (link to `docs/PREREG.md`): n per half, average R per half after costs, beats mirror and random per half, R under cost stress, and a PASS/FAIL chip. A post-hoc trial carries a POST-HOC tag. The numbers in this mock are invented.
3. **R distribution**: Histogram of trade R after costs for the selected trial, plus trades, win %, average R and the worst 3-month window (momentum crashes are the main risk).
4. **Paper vs backtest**: The parity job's result: replayed trades that match paper, fill differences, and whether paper's average R sits inside the backtest's 80% interval (PLAN P4-4).
5. **Report**: The selected trial's report: by year, India VIX tercile and results season. "Open full report" shows the whole JSON as tables.

15 · Health: is everything working, and are we ready for live?

`#/system · System › Health`

CLUBS Go-live + Health + Data + Diagnostics (crypto: System › Go-live, Health, Data, Diagnostics)

FOR When something looks wrong; weekly go-live review.

DATA `/api/golive` (60 s) · `/api/health` (30 s) · `/api/data-health` (5 min) · `/api/events` (60 s) · `/api/tables` (admin, on demand)



SAMPLE DATA, FOR LAYOUT ONLY. THE REAL PORTAL SHOWS ENGINE DATA OR AN EMPTY STATE.

1. **Verdict**: READY / NOT READY with the number of open gates. Only the server decides this; the page just shows it.
2. **Go-live gates**: Each gate with ✓/✗ and the fact behind it: lab gate passed, ≥ 6 weeks and ≥ 100 paper trades (PLAN P4-4), parity clean 2 weeks, costs verified on a real contract note, holiday file for this year verified by the owner, static IP matches (live only), square-off lead ≥ 15 minutes before the broker's cut-off, Angel login works.
3. **Engine**: Uptime, memory vs the 600 MB systemd limit (amber over 450), version and git hash, DB size, last backup.
4. **Jobs**: Today's scheduled jobs with last run, result in words and next run. On a holiday: "market closed: Diwali-Balipratipada. Only night jobs run."
5. **Sources**: Every outside source with the age of its last good answer and its state (ok / failing / proxy). This replaces the Diagnostics page.
6. **Data coverage**: Lake, corporate actions, pre-open capture days, 1m bars. Missing days are listed by date. A "show tables" link opens the newest rows of each table (admin), which replaces the Data page.
7. **Events**: The audit log: logins, settings changes, fills, exits, kill switch, alarms.

16 · Settings: change what is allowed, keys, universe

`#/system/settings · System › Settings`

CLUBS Settings + Keys + Watchlist + API list (crypto: System › Settings, Keys, API list; Command › Watchlist)

FOR Admin only (cookie). Rarely: enabling a strategy after a gate, fixing keys, excluding a stock.

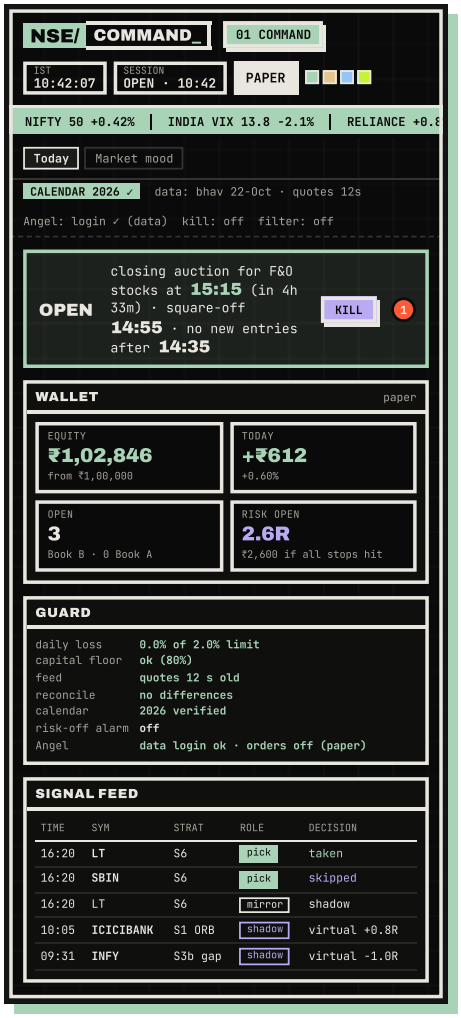
DATA `/api/portal/meta` · `POST /api/settings {key, value}` (whitelisted keys only) · `/api/keys` · `POST /api/keys/save` · `POST /api/keys/test-angel` · `/api/universe`



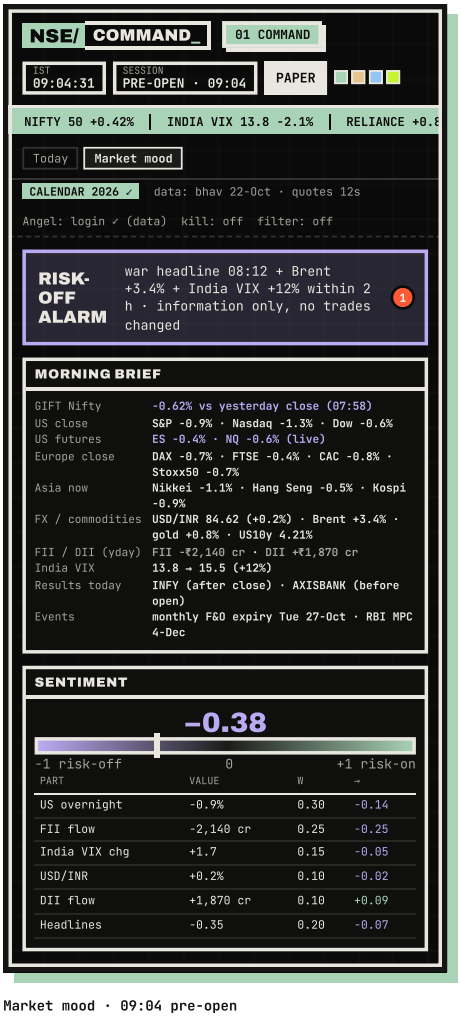
SAMPLE DATA, FOR LAYOUT ONLY. THE REAL PORTAL SHOWS ENGINE DATA OR AN EMPTY STATE.

1. **Runtime settings**: Only whitelisted keys (PLAN A6 "runtime" + this plan's new ones). **mode** and **live_orders** are never on this page: they are server .env only. A strategy or filter can only be switched on when its trial passed the gate (the control is greyed with "gate not passed" otherwise). Every change writes an event with old → new value and who.
2. **Session times**: Read-only view of today's regime from `sessions.json` and the derived square-off and no-entry times, including the next known change (7 Dec 2026).
3. **Keys**: The encrypted /keys store under the new header (V0-6, V0-8). Every key shows its **saved value masked** (first 2 + last 2; MPIN fully masked), length, when it was saved and a keyed fingerprint (none for the MPIN or any short value, because a hash of a 4-digit PIN can be reversed by trying all 10,000), so you can see at a glance what is stored. **Compare**: paste a value and the server says only "matches" or "differs at character N". **Reveal**: type the admin password again and the full current value shows for 30 s with a copy button, then hides; every reveal is logged (name only) and limited to 5 per 10 min. **History**: the last 5 saves per key, masked with fingerprints and times. Old secrets are never kept in plain text. Compare and reveal work only over HTTPS, Tailscale or localhost (greyed on plain HTTP). **Test Angel (read-only)** runs login → funds → holdings → positions → logout and explains the result in words.
4. **Universe overrides**: Manual exclude/include with a reason (stored in `config/universe_overrides.json`), next to automatic exclusions (not in Nifty 100, surveillance). Replaces the Watchlist page.
5. **API list**: Link to the endpoint list (kept as a plain server page).

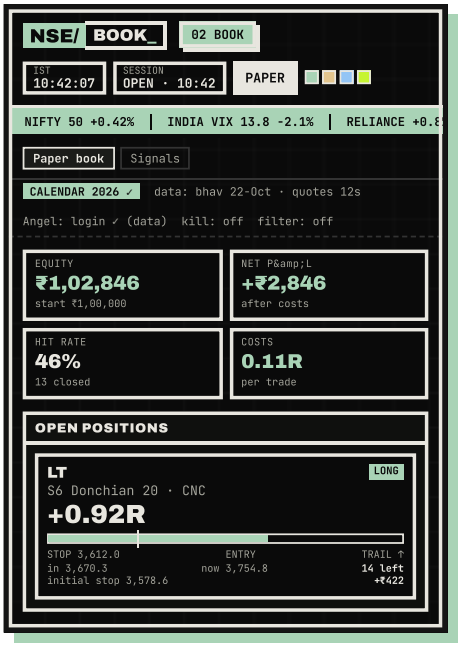
17 · Phone views (375 px; rows stack, tables scroll inside their panel)



Today · market hours



Market mood · 09:04 pre-open



Paper book · evening

On a phone only the active hub tab shows in the header; the others sit behind the logo menu. Panels keep their order from the desktop page. Wide tables become horizontally scrollable inside their panel, with the first column pinned.

18 · Telegram messages (separate NSE bot; quiet 22:00–07:30 except alarms)

WHEN	MESSAGE	EXAMPLE
08:45 trading day	Morning brief (same as Market mood)	`NSE · Fri 23-Oct brief: GIFT Nifty -0.6% · US S&P -0.9% · Europe DAX -0.7% · Asia Nikkei -1.1% · Brent +3.4% · USD/INR 84.62 · FII -₹2,140 cr / DII +₹1,870 cr · VIX 13.8 · mood -0.38 (risk-off) · results: INFY, AXISBANK`
09:20	Opening summary	`Open: Nifty -0.71% (pre-open imbalance -0.18 sell-heavy) · breadth 31/69 · first 5 min vol 1.8× · gap not filled`
any time	Risk-off alarm (max 1/day)	`RISK-OFF: war headline 08:12 + Brent +3.4% + VIX +12% in 2 h. Information only, no trades changed.`
16:15	Fill / exit	`PAPER FILL LT BUY 5 @ 3,670.30 · stop 3,578.60 · risk ₹458` · `PAPER EXIT INFY stop (gap open) @ 1,468.0 · -1.32R · -₹1,330 after ₹164 costs`
16:20	Orders prepared	`Tomorrow 09:15: LT LIMIT 3,670.30 ×5 · TITAN LIMIT 3,402.10 ×6 · skipped SBIN (position under ₹20,000)`
16:35	Daily summary	`Day: +₹612 (+0.60%) · equity ₹1,02,846 · 3 open (2.6R at risk) · costs today ₹168 · signals 2 taken / 1 skipped`
on failure	Errors (immediately)	`Angel login failed (AB1050): TOTP doesn't match client. Paper continues on bhavcopy.` · `No 2027 holiday file: engine won't trade from 1 Jan.` · `Feed stale 60 s` · `Kill switch ON (portal, owner)` · `Parity fault 2 days in a row`
commands	From the owner's chat id only	`/status` · `/positions` · `/brief` · `/kill` · `/resume` · `/totp 123456` (fallback login)

Never in a message: keys, tokens, MPIN, TOTP, or the full client code.

19 · Angel API check (23 Oct 2026, read-only)

Run on the EC2 (read-only, script piped over SSH, nothing written to disk, keys read in a read-only Postgres transaction). No orders, no tokens printed.

TIME (IST)	RESULT	MEANING
17:10	login refused · AB1050 "Invalid totp and client combination"	secret valid base32 (26 chars), server clock +0.5 s, so the secret didn't belong to this client
17:20	no login attempted	newly saved secret had a symbol in it (20 chars), not valid base32
17:26	login refused · AB1050	new 26-char secret (saved 17:24) is valid base32 but still not paired with client AA***14

TO FIX (OWNER)

1. Open `smartapi.angelone.in/enable-totp` and log in with **the same client code** saved on /keys (your Angel client ID, not mobile or email), MPIN and the OTP.
2. Copy the **text secret** shown under the QR code (letters A–Z and digits 2–7, about 26–32 characters, no spaces). Not the 6-digit code.
3. Save it on `/keys → Angel TOTP secret` and **don't open the enable-TOTP page again** afterwards (it can issue a new secret and cancel the saved one).
4. Check that the SmartAPI app (API key) was created under the same client code.
5. Ask Claude to rerun the read-only check (one attempt), or use the **Test Angel** button once V0-6 is built.

Once login works, the same check prints available cash, margin used, and every holding (symbol, qty, average, LTP, P&L), then logs out.

Appendix A · Full build plan (docs/PLAN_V2.md)

Written 23 Oct 2026 by Claude (planner) for Cursor (builder). The owner's brief: "make it similar in UI, paper trading and basic concept; only market-related things change", and "Claude plans everything in detail, Cursor does the hard work".

How to use this file (Cursor): read `AGENTS.md`, then this file, then `docs/MARKET_INDIA.md` (the market facts this plan relies on). `docs/PLAN.md` still holds, except where this file says otherwise. Where they disagree, this file wins (owner decision, logged in `docs/DECISIONS.md` on 23 Oct 2026). Work one phase per branch, one PR per phase. Every ticket has a test that fails before and passes after. Ask the owner, don't guess, when §11 lists the question as open. Use the default written there until answered.

The reference implementation is the owner's crypto engine on this Mac: `/Users/dhruv/personal/Tennis-bet/crypto-signal-engine` (public repo). Copy its front-end and ideas freely into this private repo. Never copy anything from this repo into that one.

1. What "similar to crypto" means

The crypto engine reached its current shape through a lot of evidence: one swing book that is allowed to paper trade, intraday quarantined to shadow, every signal logged with mirror and random baselines, every signal replayed to an outcome (traded or not), filters audited as "taken vs skipped", a wallet-based paper cycle, a go-live checklist, and a four-hub "Brutal Command" portal. The NSE engine copies all of that. Only the market layer is different.

CONCEPT	CRYPTO ENGINE (REFERENCE)	NSE ENGINE (TARGET)
The book that paper trades	Swing book: 4h/8h breakouts, stop 3×ATR, target 3R, 7-day limit (`analysis/swing_book.py`)	**Book B**: daily breakouts S6/S6b (PLAN A10), CNC, long only, stop 2.5×ATR14 trailed, 20-session limit
Quarantined family	15m detectors run shadow only	**Book A** (S1–S5 intraday MIS) runs **shadow only**: signals and virtual outcomes, no paper trades, until it passes the gate (§2 D2)
Market filter	Skip swing signals when BTC 30-day vol is in the top third of its past year (`analysis/regime_gate.py`)	Same idea on **Nifty 50** 30-day realised vol, as a pre-registered **trial** first (V3), not switched on by assumption
Paper wallet	Paper cycle: starting wallet, target wallet, peak, equity, unrealised, hit rate, net P&L, costs (`analysis/paper_cycle.py` , Book › Paper cycle)	Same cycle object in ₹. Costs split into brokerage, STT, exchange, SEBI, stamp, GST, DP, slippage
Fill rule shared by backtest and paper	`resolve_candle` / `virtual_outcome`	New `core/exits.py` , used by the backtest, the paper engine and the outcome replay (V2-7)
Every signal resolved	`_swing_outcomes_job` replays traded and skipped signals	`job_outcomes` replays every pick, mirror, random and skipped signal on daily bars
Costs	Fees + 18% GST + slippage + funding	STT, stamp, exchange, SEBI, GST, DP, brokerage with min ₹5, slippage (`MARKET_INDIA.md` §5)
Sessions	24/7	Pre-open, continuous, closing auction, holidays, Muhurat. Times are dated and per stock class (`MARKET_INDIA.md` §2)
Leverage	Up to 10x	None for CNC. MIS ≤5x (shadow only for now)
Shorting	Either side	CNC long only. MIS either side
Settlement	Instant	T+1. Book B holds at least one night
Data	Binance klines + WS	NSE bhavcopy (UDiFF) + Angel SmartAPI candles, quotes and WS. Yahoo only for the ticker tape
Portal	4 hubs: Command / Book / Evidence / System; micro-UI `core.js` ; 4 palettes	Same hubs, pages, components and palettes; NSE data
Go-live	System › Go-live gates page	Same page, with NSE gates (§8)
Alerts	Telegram	Telegram, separate bot (unchanged)

2. Decisions this plan proposes (owner confirms in §11)

- **D1. Book B paper trades first, on end-of-day data.** Signals are computed on the daily close and filled the next day by exactly the backtest's daily fill rule. Paper doesn't depend on a live tick feed, and paper equals backtest by construction (crypto lesson 4).
- **D2. Book A runs shadow only** (like crypto's 15m detectors). PLAN A2's own numbers mean Book A can't size a trade at ₹50k: the ₹1L minimum with 1% risk needs a stop of 0.5% or less, which costs 0.29R. It runs on the live Angel feed in V6, logs signals and virtual outcomes, and is promoted only through the gate.
- **D3. Paper capital ₹1,00,000** with 1% risk. That gives ₹20k-plus CNC positions at about a 5% stop. At ₹50k and 5 slots each position is ₹10k, and costs reach 0.79% a round trip (`MARKET_INDIA.md` §5). New setting `min_position_cnc_inr = 20000` .
- **D4. Square off ≥15 minutes before the broker's cut-off,** derived from one setting, not hard-coded. CAS moved brokers' cut-offs to about 15:00–15:10 for F&O stocks until 6 Dec 2026.
- **D5. Lab first, then filters.** The Nifty vol filter, a trend filter and one momentum strategy are added as pre-registered trials (V3). Nothing is switched on without passing the gate.
- **D6. One fill/exit engine** (`core/exits.py`) for backtest, paper and outcome replay.
- **D7. The portal is a port of crypto's micro-UI** (Store, widgets, views, components, palettes), with NSE data behind the same kind of JSON endpoints.
- **D8. Track the whole opening and the market's mood every day, from as early as possible** (owner, 23 Oct 2026: "market opening is a very big thing, so track entire sentiment with GIFT Nifty, US and Europe market, war news, every major thing"). Record global cues, GIFT Nifty, FII/DII, India VIX, tagged major-event headlines, the 09:00–09:15 pre-open auction and the first 30 minutes. Pre-open and live breadth can't be downloaded later, so this is phase **V1b**, right after V1. It is **information and research data**: it never blocks or sizes a trade until a pre-registered lab trial shows lift (crypto: news echoed price 48/48).

3. Review: where the app is today

3.1 WHAT WORKS

- Settings, the dual live flags and the pytest guard. Live order methods are hard-gated (`broker/angel.py:107-110`).
- Cost model formulas (`core/costs.py`) match Angel's structure. Only the rates need refreshing (V1-4).
- Indicators (SMA, EMA, Wilder ATR and RSI, VWAP, Donchian, Keltner), SQLite migrations, ledger, keys page with an encrypted store, Telegram client, market-hours deploy guard.
- Tests: 54 pass. Ruff has 2 errors (unused imports in `scripts/_remote_check_keys_quotes.py:6` and `scripts/login_check.py:19`), so CI is red.

3.2 WHAT ISN'T THERE YET

- **No paper trading runs.** Every trading job in `engine/runner.py:34-59` only logs. `/api/today` returns zeros (`web/portal_api.py:29-33`). `/api/health` hard-codes `login_ok: false` (`web/app.py:38`).
- **All research data is synthetic.** Every file in `data/lake/bars_1d/` has `source == "synth"` . The S6 report (45 trades, −0.26R) is a smoke test, not evidence.
- **The portal is a skin with placeholders.** It has the crypto look and four hubs, but none of crypto's pages (swing positions in R, paper cycle tiles, signals with "why refused", taken vs skipped, go-live gates, session guard, accuracy, history), one palette instead of four, and no component kit.

- **Nothing is committed.** The whole tree is untracked on `feat/mvp-backtest-news` (only "Initial commit" exists).

3.3 BUGS (CONFIRMED BY RUNNING PROBES ON 23 OCT 2026 UNLESS MARKED "BY READING")

#	SEVERITY	WHERE	WHAT IS WRONG	FIXED IN
B1	CRITICAL	`lab/backtest.py:189-197`	Intraday backtest checks the stop **only for longs**. A gap-fade short with its stop at 103.2 ran to 120.6: **−30R** recorded as one "square_off" trade	V2-2
B2	HIGH	`lab/backtest.py:157-230`	Intraday targets and `time_exit_ms` are ignored. Square-off happens at the last bar (15:29 in the probe), not 15:05. An ORB trade that reached 2R was held to the close	V2-2
B3	CRITICAL	`lab/backtest.py:78-83`	Daily stops always fill **at the stop price, even when the day opens far below it**. Probe: a −20% gap recorded −1.46R where the real fill (at the open) is about −10.6R. Backtests look far safer than reality	V2-1
B4	HIGH	`lab/backtest.py:114-124`	Daily entry assumes the next-day LIMIT (close +0.5%) always fills at that price. No check that the next day traded down to it, no better fill when it opens below, `entry_ms` set on the signal day. The trailing stop (PLAN A10) is missing	V2-1
B5	CRITICAL	`scripts/research.py:38-73`	**Mirror and random baselines are made up**: mirror R = −pick R, random R = 0. The gate's "beats mirror / beats random" checks reduce to "average R > 0". This breaks crypto lesson 2	V2-4
B6	HIGH	`scripts/research.py`, `lab/backtest.py:49`	Research runs one symbol at a time with its own capital. The n ≥ 100 per half gate can never be met, and there's no portfolio (shared cash, max open, priority)	V2-5
B7	CRITICAL	`broker/paper.py:154-161`	Paper MIS close returns only the margin. **Profit and loss never reach cash** (probe: +₹10,000 trade, cash unchanged at ₹1,00,000). Costs are never deducted. Partial reduces don't touch cash. Delivery buys can push cash negative	V4-3
B8	HIGH	`core/orders.py:239`	The stop watchdog measures the stop order's age, not how long price has been through the trigger. It forces an exit on the **first** crossing tick if the stop is more than 10 s old (probe confirmed). PLAN A9.4 says "crossed for longer than `stop_watchdog_s`"	V4-5
B9	HIGH	`core/orders.py:92,95,140,142,255`	Prices are rounded to 2 decimals, not the tick (probe: 1235.18 and 1216.11, invalid on a ₹0.10 tick). Breaks AGENTS rule 8. The exchange rejects such orders	V0-4
B10	MEDIUM	`core/orders.py:96-100`	"Idempotent by tag" can't work: a new tag is generated, then looked up	V4-5
B11	HIGH	`core/orders.py:114` vs `broker/angel.py:112,128`	Order manager calls `place_order(**kwargs)` / `cancel_order(client_tag)`. Angel takes `place_order(params: dict)` / `cancel_order(order_id, variety)`. `broker/base.py` is an empty async stub while both brokers are sync. "Paper uses exactly the live code path" isn't true yet	V4-5
B12	MEDIUM	`core/orders.py:268-276`	`square_off` cancels the stop and target but sends no exit	V4-5
B13	HIGH (BY READING)	`strategies/base.py:23-24`	`BarView.closed()` treats every bar as closed 60 s after its open. A daily bar stamped 09:15 counts as closed at 09:16, so a daily strategy run live would see the day's high, low and close hours early	V2-3
B14	HIGH (BY READING)	`strategies/orb.py:43-50`, `vwap_revert.py`, `intraday_momentum.py`	Strategies are fed 1m bars, but the rules are 5m/15m. ORB uses a 1m close and 1m volume against a 5m slot average. ATR is on 1m bars where PLAN says 5m	V2-2
B15	CRITICAL (BY READING)	`strategies/gap.py`, `pdh_rs.py`, `intraday_momentum.py`, `orb.py`	**Nothing ever supplies `prev_close`, `pdh`, `rs_rank`, `ret_decile`, `slot_vol_avg` or `gap_pct`** (no caller in `lab/`, `engine/`, `scripts/` or `tests/`). S3, S3b, S4 and S5 can never fire. S1's volume filter is silently skipped, so it isn't the pre-registered rule	V2-2
B16	HIGH	`core/clock.py:55-102`	One fixed 15:30 close for every stock. Since 3 Aug 2026, F&O stocks stop continuous trading at 15:15 (CAS), and from 7 Dec 2026 at 15:30 with CAS to 15:40. No pre-open phases	V1-1
B17	HIGH	`config/holidays_2026.json:23-32`	Muhurat placeholder 18:15–19:15. The real 2026 session is in the **afternoon** (14:20 start)	V1-2
B18	HIGH	`data/calendar.py:77-80`	Only the 2026 file loads. From 1 Jan 2027 every weekday counts as a trading day. The runner's cron (`engine/runner.py:93-105`) runs Mon–Fri jobs on holidays too	V1-2
B19	HIGH	`settings.py:61-62`, PLAN A3	Square-off 15:05 can be later than Angel's post-CAS cut-off (reported 15:10 or earlier for F&O stocks)	V1-3
B20	MEDIUM	`config/costs_angel.json:10`	NSE transaction charge 0.00297%. Current tables show 0.00307%. Brokerage minimum ₹5 not modelled. `verified_against_calculator: false`	V1-4
B21	HIGH	`data/history.py:23-50`	Bhavcopy parser reads only the legacy columns. NSE's file has been UDiFF since July 2024. No split/bonus/dividend adjustment anywhere, so a 1:1 bonus looks like a −50% gap	V1-7
B22	MEDIUM	`data/free_quotes.py:22`	"Live" quotes are Yahoo daily candles (`interval=1d&range=5d`), delayed and unofficial, for 20 symbols. Fine for the tape, unusable for fills	V4-2
B23	MEDIUM	`scripts/research.py:25`, `engine/runner.py:26`	Research writes to `./data/engine.db`. The portal reads `./var/engine.db` (`settings.data_dir`), so research trials never show on the Lab page	V0-5
B24	HIGH	`config/settings.py:76`, DECISIONS 23 Oct	Portal binds `0.0.0.0` and is served publicly over **plain HTTP** (`http://52.62.37.4/nse/`). The admin password and Angel keys typed on `/keys` cross the internet unencrypted. Cookie has no `Secure` flag. PLAN A13 said Tailscale only	V0-2
B25	LOW	`core/orders.py:87,148` and similar	`now_ms or time.time()` treats 0 as missing (a test at ms=0 silently uses wall time)	V0-3
B26	LOW	`core/risk.py:104-107`	MIS exposure check ignores the new position's own value	V1-5
B27	LOW	`engine/runner.py:61-78`	News job ingests an empty list (never fetches). Fine while news is off. Leave it off	—

4. Target architecture (what changes, file by file)

```
config/
  sessions.json            NEW  dated session regimes per stock class (V1-1)
  holidays_2026.json       FIX  Muhurat afternoon; schema adds "verified_by_owner" date (V1-2)
  holidays_2027.json       NEW  when NSE publishes (owner), loader requires it from 1 Jan 2027
  costs_angel.json         FIX  rates, min brokerage, source + date, verified_against_contract_note (V1-4)
  tick_bands.json          NEW  fallback tick table (MARKET_INDIA §4) (V0-4)
  universe_history.csv     NEW  Nifty 100 adds/removes with dates (V1-6)
  surveillance.json        NEW  ASM/GSM exclusions, refreshed daily (V1-5)
core/
  ticks.py                 NEW  round_to_tick(price, tick, direction) (V0-4)
  clock.py                 FIX  session_state(ms, *, stock_class) reads sessions.json (V1-1)
  exits.py                 NEW  the one fill/exit engine (V2-7)
  costs.py                 FIX  min brokerage; breakdown incl. slippage (V1-4)
  risk.py                  FIX  min_position_cnc_inr, exposure incl. new position, surveillance (V1-5)
  orders.py                FIX  tick rounding, real idempotency, watchdog timing, square-off exit (V4-5)
broker/
  base.py                  FIX  real sync Broker protocol + dataclasses (Order, Fill, Position, Funds, Quote) (V4-5)
  paper.py                 FIX  cash/P&L/costs accounting; CNC long-only; T+1 holding rule (V4-3)
  angel.py                 FIX  adapter to the Broker protocol (tag → order id map) (V4-5)
data/
  history.py               FIX  UDiFF + legacy bhavcopy, delivery file, corporate actions (V1-7)
  corporate_actions.py     NEW  adjustment factors (V1-7)
  events.py                NEW  results dates, expiry Tuesdays, RBI MPC, Budget (V1-8)
  quotes.py                NEW  Angel LTP poller for marking; Yahoo fallback for the tape (V4-2)
  global_cues.py           NEW  US / Europe / Asia indices + futures, USD/INR, DXY, Brent, gold, US 10y, GIFT Nifty (V1b-1)
  preopen.py               NEW  NSE pre-open snapshots (IEP, buy/sell qty) every 30 s, 09:00-09:15 (V1b-2)
  flows.py                 NEW  FII/DII provisional cash flows each evening (V1b-1)
news/
  india_rules.py           FIX  new event tags: geopolitics_india, middle_east, russia_ukraine, tariffs_trade, us_data,
                                china, opec, monsoon, gst_council, india_macro, block_deal, promoter_pledge (V1b-3)
  scrape.py                FIX  real fetch on a schedule, more feeds, published time from the feed (V1b-3)
engine/
  opening.py               NEW  morning brief, opening scorecard, opening type label, risk-off alarm (V1b-4)
lab/
  backtest.py              REWRITE on core/exits.py; portfolio mode across the universe (V2)
  baselines.py             NEW  real mirror + random trades (V2-4)
  evaluate.py              FIX  halves, by year, VIX tercile, results season, cost stress (V2-6)
  prereg.py + docs/PREREG.md NEW  trial definitions written before runs (V3-2)
engine/
  paper_engine.py          NEW  Book B EOD cycle: scan → prepare → fill → exits → mark (V4)
  outcomes.py              NEW  resolve every signal (taken/skipped/mirror/random) (V4-4)
  runner.py                FIX  real jobs, calendar-aware (V4-6)
  migrations/002_paper.sql NEW  §5
web/
  portal/ (core.js, portal.css, portal.html, nav.js, hubs/*.js)   PORT from crypto (V5)
  portal_api.py            FIX  the endpoints in §7
```

5. Data model changes (`engine/migrations/002_paper.sql`)

```
CREATE TABLE paper_cycles (
  id INTEGER PRIMARY KEY, started_ms INTEGER NOT NULL, ended_ms INTEGER,
  starting_wallet REAL NOT NULL, target_wallet REAL, wallet REAL NOT NULL, peak_wallet REAL NOT NULL,
  status TEXT NOT NULL DEFAULT 'running',          -- running | target_hit | floor_hit | stopped
  outcome TEXT, config_json TEXT NOT NULL           -- the exact rules in force when the cycle started
);
ALTER TABLE trades ADD COLUMN cycle_id INTEGER;
ALTER TABLE trades ADD COLUMN costs_json TEXT;      -- {brokerage, stt, exchange, sebi, stamp, gst, dp, slippage}
ALTER TABLE trades ADD COLUMN mfe_r REAL;           -- best R reached
ALTER TABLE trades ADD COLUMN mae_r REAL;           -- worst R reached
ALTER TABLE trades ADD COLUMN sessions_held INTEGER;
ALTER TABLE trades ADD COLUMN note TEXT;
ALTER TABLE signals ADD COLUMN virtual_status TEXT; -- running | resolved | no_fill
ALTER TABLE signals ADD COLUMN virtual_exit_reason TEXT;
ALTER TABLE signals ADD COLUMN virtual_r REAL;      -- after costs, same exit engine as real trades
ALTER TABLE signals ADD COLUMN resolved_ms INTEGER;
ALTER TABLE signals ADD COLUMN filter_tag TEXT;     -- e.g. "nifty_vol_rank=0.81" (crypto regime tag)
CREATE TABLE pending_orders (                       -- Book B entries prepared in the evening for tomorrow
  id INTEGER PRIMARY KEY, signal_id INTEGER NOT NULL, trade_date TEXT NOT NULL, symbol TEXT NOT NULL,
  side INTEGER NOT NULL, qty INTEGER NOT NULL, limit_price REAL NOT NULL, stop REAL NOT NULL,
  status TEXT NOT NULL DEFAULT 'pending', created_ms INTEGER NOT NULL
);
```

`lab_trials` also gets `prereg_id TEXT`, `data_hash TEXT` and `post_hoc INTEGER DEFAULT 0` (today `post_hoc` is hidden inside `params_json`, `lab/ledger.py:33-35`).

Opening and sentiment tables (`engine/migrations/003_opening.sql`, phase V1b):

```
CREATE TABLE global_cues (            -- one row per instrument per snapshot
  ts_ms INTEGER NOT NULL, key TEXT NOT NULL,          -- "SPX", "NDX", "ES", "DAX", "FTSE", "N225", "USDINR", "BRENT", "GIFTNIFTY" ...
  last REAL, chg_pct REAL, ref_close REAL, source TEXT NOT NULL, stale INTEGER DEFAULT 0,
  PRIMARY KEY (ts_ms, key)
);
CREATE TABLE flows (trade_date TEXT PRIMARY KEY, fii_net_cr REAL, dii_net_cr REAL, source TEXT, final INTEGER DEFAULT 0);
CREATE TABLE headlines (
  id INTEGER PRIMARY KEY, ts_ms INTEGER NOT NULL, published_ms INTEGER, source TEXT, title TEXT NOT NULL, link TEXT,
  event_type TEXT, symbol TEXT, score REAL, confidence REAL, market TEXT, external_id TEXT UNIQUE
);
CREATE TABLE preopen_snapshots (
  trade_date TEXT NOT NULL, ts_ms INTEGER NOT NULL, symbol TEXT NOT NULL,   -- "NIFTY" for the index
  iep REAL, prev_close REAL, chg_pct REAL, buy_qty INTEGER, sell_qty INTEGER, final INTEGER DEFAULT 0, source TEXT,
  PRIMARY KEY (trade_date, ts_ms, symbol)
);
CREATE TABLE opening_summary (
  trade_date TEXT NOT NULL, symbol TEXT NOT NULL, open REAL, prev_close REAL, gap_pct REAL,
  imbalance REAL,                                 -- (buy_qty - sell_qty) / (buy_qty + sell_qty) at the final pre-open
  or15_high REAL, or15_low REAL, or30_high REAL, or30_low REAL, first5_vol_x REAL,
  gap_filled_ms INTEGER, close REAL, open_to_close_pct REAL, open_type TEXT,
  PRIMARY KEY (trade_date, symbol)
);
CREATE TABLE morning_brief (trade_date TEXT PRIMARY KEY, built_ms INTEGER, json TEXT NOT NULL);  -- frozen at 08:45
CREATE TABLE sentiment_daily (
  trade_date TEXT PRIMARY KEY, composite REAL, parts_json TEXT NOT NULL,   -- each component, its weight, its value
  breadth_adv INTEGER, breadth_dec INTEGER, risk_off_alarm INTEGER DEFAULT 0
);
```

6. New and changed settings (`config/settings.py`; never touch `mode` / `live_orders` defaults)

NAME	DEFAULT	WHY
`capital_inr`	100000	D3 (paper capital; live reads funds from Angel)
`min_position_cnc_inr`	20000	Book B cost floor (MARKET_INDIA §5)
`book_a_mode`	"shadow"	D2. Values: off / shadow / paper
`broker_squareoff_at`	"15:10"	D4. Owner confirms Angel's cut-off (Q1)
`squareoff_lead_min`	15	engine square-off = broker − lead
`no_entry_lead_min`	20	no new entries = engine square-off − lead
`paper_target_wallet`	0 (none)	crypto-style cycle target, optional
`market_filter`	"off"	"off" / "nifty_vol" / "nifty_trend". On only after the V3 gate
`market_filter_vol_rank_max`	0.667	crypto's "top third" rule
`web_host`	"127.0.0.1"	B24
`cookie_secure`	true	B24 (false only for local http dev)
`opening_tracker_enabled`	true	D8. Recording only, never a trade filter
`preopen_snapshot_s`	30	pre-open poll interval
`cues_poll_s`	300	global cues every 5 min, 06:00–23:30 IST
`risk_off_brent_pct` / `risk_off_vix_pct` / `risk_off_window_min`	3.0 / 10.0 / 120	risk-off alarm rule (MARKET_INDIA §8b)
`gift_nifty_source`	"none"	"none" / "url" / "paid". Until set, US futures + Asia are the overnight proxy (Q11)

`square_off_at` and `no_entry_after` become derived values, shown on the portal. Tests assert the engine square-off is always at least `squareoff_lead_min` before `broker_squareoff_at`.

7. API contract for the portal (JSON shapes the front end codes against)

The front end is built against these shapes. V5 can start with fixtures in `tests/fixtures/portal/*.json` before the engine fills them. All times are UTC epoch ms; the browser shows IST.

```
GET /api/portal/meta     {mode, live_orders, kill_switch, book_a_mode, market_filter,
                          session: {state, stock_class, label, next_change_ms, next_label},
                          calendar_ok, calendar_year_loaded, costs_verified, capital_inr,
                          squareoff_at, no_entry_after, broker_squareoff_at, version}
GET /api/quotes          {ts_ms, source, quotes: [{symbol, ltp, prev_close, chg_pct, volume}]}   (tape)
GET /api/market          {nifty: {ltp, chg_pct}, india_vix, nifty_vol_rank, nifty_above_200dma,
                          fii_net_cr, dii_net_cr, gift_nifty_chg_pct|null, next_events: [{kind, date, label}]}
GET /api/swing           {rules: {timeframe, stop, trail, time_limit_sessions, risk_pct, min_position_inr,
                                  max_open, market_filter, excluded},
                          open: [{trade_id, symbol, strategy, qty, entry, stop_init, stop_now, mark,
                                  r_now, unrealised_inr, sessions_held, sessions_left, entry_ms}],
                          closed: {n, win_rate, avg_r, total_r, last: [{symbol, strategy, r, net_inr,
                                   costs_inr, exit_reason, entry_ms, exit_ms}]},
                          takeskip: {take: {n, running, avg_r}, skip: {n, running, avg_r}}}
GET /api/paper           {enabled, running,
                          cycle: {id, started_ms, starting_wallet, target_wallet, wallet, peak_wallet},
                          equity, unrealised, positions: [...same as swing.open + book],
                          summary: {trades, win_rate_pct, net_pnl, expectancy_per_trade,
                                    costs: {brokerage, stt, exchange, sebi, stamp, gst, dp, slippage}},
                          history: [{...closed trade, book}]}
GET /api/portal/signals?days=14&book=&strategy=&role=&decision=
                         {rows: [{id, ts_ms, symbol, book, strategy, role, side, price, stop, target,
                                  decision, skip_reason, reason_text, virtual_status, virtual_r}],
                          refused: [{reason, reason_text, n}]}
GET /api/lab             {trials_total, strategies: [{id, book, last_report: {gate fields...}, trials,
                          passed, prereg_id}]}
GET /api/golive          {verdict, gates: [{id, label, ok, detail}]}
GET /api/guard           {session, feed: {last_tick_age_s, stale}, reconcile_faults: [...],
                          daily_loss_pct, floor_ok, kill_switch, flags: [{id, on, text}]}
GET /api/data-health     {lake: {symbols, first_date, last_date, missing_days: [...]},
                          bhav: {last_date, format}, corp_actions_applied, minutes_today: {...}}
GET /api/opening         {trade_date, phase: "overnight"|"preopen"|"opening"|"day"|"closed",
                          brief: {gift_nifty: {chg_pct, ts_ms}|null, us: [...], europe: [...], asia: [...],
                                  fx_commod: [...], fii_net_cr, dii_net_cr, india_vix, top_headlines: [...],
                                  results_today: [...], events_today: [...], composite, built_ms},
                          preopen: {nifty: {iep, chg_pct, imbalance}, snapshots_n, last_ms,
                                    movers: [{symbol, iep, chg_pct, imbalance}]},
                          scorecard: {nifty_gap_pct, gap_filled_ms|null, or15: [lo, hi], breadth: {adv, dec},
                                      first5_vol_x, open_type|null},
                          history: [{trade_date, nifty_gap_pct, imbalance, open_type, open_to_close_pct}]}
GET /api/sentiment       {composite, label, parts: [{key, label, value, weight, contribution, ts_ms, stale}],
                          markets: [{key, label, region, last, chg_pct, ts_ms, stale}],
                          risk_off_alarm: {on, since_ms, reasons: [...]},
                          timeline: [{ts_ms, nifty_chg_pct, composite}]}
GET /api/headlines?hours=24&tag=&symbol=
                         {rows: [{ts_ms, source, title, link, event_type, symbol, score, nifty_move_next_60m|null}],
                          tags: [{event_type, n}]}
GET  /api/keys           (admin) {rows: [{key, label, kind, is_set, where, masked, length, fingerprint, saved_ms,
                          history: [{masked, fingerprint, saved_ms, saved_by}]}], secure_channel: bool}
POST /api/keys/compare   (admin, secure channel) {key, value} → {match: bool, first_diff_at: int|null}
POST /api/keys/reveal    (admin + password, secure channel) {key, password} → {value}   no-store, audited, 5 per 10 min
GET /api/events?limit=   audit events
GET /api/tables          (admin) newest rows per table, like crypto's System › Data
```

`reason_text` maps every machine reason to plain English in one dict (`web/reasons.py`). For example, `below_min_position` → "position under ₹20,000: costs would eat too much", `near_circuit` → "too close to today's price band", `results_day` → "results due within 3 days", `market_filter` → "Nifty volatility in the top third of the past year".

8. Portal: pages and widgets (port of crypto `scheduler/portal/`)

Keep hash routing (`#/book/paper`) because the portal is served under `/nse/` . Everything else follows `crypto/docs/PORTAL_REVAMP.md` : the Store (poll only while a widget uses a key, pause in hidden tabs), widgets, views, the component kit `UI.C` , the four palettes (Sage & Lilac default, Champagne & Teal, Ice & Peach, Classic lime) with header swatches and `localStorage` , motion only as decoration, `prefers-reduced-motion` , and **nothing simulated on the real portal**.

Seven pages, not twenty-one (owner, 23 Oct 2026: "cut them short to only required, or club them"). The four crypto hubs stay, so the portal looks and navigates like crypto. Each hub holds only the pages that earn their place. Anything that was only another view of the same data is now a section or filter on one page.

HUB	PAGE (ROUTE)	WHAT IT CLUBS	SECTIONS, TOP TO BOTTOM
01 Command	**Today** (`#/command`)	Overview + Session guard + Live ₹ status	session hero (state, countdown to next change, square-off and no-entry times, kill switch); pipeline (data → strategies → risk → paper book → opening recorder); wallet mini-tiles (equity, today's P&L, open positions); guard flags (daily loss, floor, feed stale, reconcile, calendar, risk-off alarm); universe heat grid; decision trace; latest signal feed + toast
	Market mood (`#/command/mood`)	Opening + Sentiment + market context	morning brief (frozen 08:45); pre-open board (09:00–09:15 live); opening scorecard (from 09:15) and opening type (15:45); markets grid (GIFT Nifty, US, Europe, Asia, FX, crude, gold, bonds, India VIX, FII/DII); sentiment gauge with parts; risk-off banner; headlines timeline with tag chips; 60-day opening history
02 Book	**Paper book** (`#/book`)	Swing book + Paper cycle + Live ₹ (locked)	cycle tiles (equity, cash, unrealised, net P&L, hit rate, costs split, cycle #); equity curve; open positions as R cards (Book B; Book A appears only if promoted); tomorrow's pending orders; closed trades with book/strategy filters and totals; taken vs skipped; rules in force; a locked "Live ₹" strip showing which gates are still open
	Signals (`#/book/signals`)	Signals + Intraday shadow + History	filter bar (date range, book, strategy, role pick/mirror/random, decision taken/skipped/shadow, symbol search); summary tiles; one table for every signal incl. Book A shadow with virtual outcome; "why refused" side panel in plain words; CSV export (this replaces the History archive)
03 Evidence	**Evidence** (`#/evidence`)	Strategies + Accuracy + Research	gate table per trial (pass/fail chips, prereg link, trials counter); R histogram and win rate by strategy/month; paper-vs-backtest check; latest research reports with by-year, VIX tercile and results-season tables
04 System	**Health** (`#/system`)	Go-live + Health + Data + Diagnostics	go-live verdict and gates; jobs (last run, result, next run); data coverage (lake, bhavcopy, corporate actions, pre-open capture days, missing days); feed/source states with age and errors; memory vs 600 MB; recent events
	Settings (`#/system/settings`)	Settings + Keys + Watchlist + API list	runtime settings (admin, whitelisted, every change logged); derived session times (read-only); universe and manual exclude/include; keys (encrypted, set / not set) with the **Test Angel (read-only)** button; API list link

Retired as pages (still reachable as sections or filters): Overview, Opening, Sentiment, Watchlist, Swing book, Paper cycle, Intraday (shadow), Live ₹, Session guard, Strategies, Accuracy, History, Research, Go-live, Data, Diagnostics, Keys, API list. Old crypto-style addresses (`#/book/paper` , `#/command/signals` , `#/system/golive` …) redirect to the page and section that now hold them.

Copy `crypto/tests/test_portal.py` 's structural checks: every native page has a view, every view names existing widgets, every widget uses a defined Store key.

9. Phases and tickets

Each phase: branch `feat/v<N>-<topic>` , PR, `ruff check .` and `pytest -q` green, `docs/phase-reports/V<N>.md` . Order: V0 → V1 → **V1b** → V2 → V3 → V4 → V6 → V7. **V5 (portal) can run in parallel with V1–V4** against the §7 fixtures. V1b goes early because pre-open and breadth history can't be downloaded later.

V0: HYGIENE AND SAFETY (`FEAT/V0-HYGIENE`)

TICKET	WORK	DONE WHEN
V0-1	Commit the current tree on `feat/mvp-backtest-news` (owner OK), fix the 2 ruff errors, open the MVP PR	CI green; `git status` clean
V0-2	B24: default `web_host=127.0.0.1` ; cookie `Secure` unless `cookie_secure=false` ; `/keys` save refuses unless the request is HTTPS (`X-Forwarded-Proto: https`) or from 127.0.0.1 / Tailscale 100.64.0.0/10. Docs: nginx TLS or `tailscale serve` (owner picks, Q2)	tests: plain-HTTP public request to `/api/keys/save` gets 403; cookie has `Secure`
V0-3	B25: replace `x or now()` with `x if x is not None else now()` everywhere	test at `now_ms=0` uses 0
V0-4	B9: `core/ticks.py` `round_to_tick(price, tick, "up"|"down"|"nearest")` with `config/tick_bands.json` fallback; every price sent to a broker goes through it. Buy limits round up, sell limits round down, stop triggers round toward the market	tests: 1234.56 on a ₹0.10 tick → 1234.60 up / 1234.50 down; 249.99 uses ₹0.01; OrderManager emits only on-tick prices
V0-6	Keys page: **"Test Angel (read-only)"** button → `POST /api/keys/test-angel` (admin): login, `rmsLimit` , `allholding` , `position` , logout; returns status, error code with a plain-English hint (AB1050 → "TOTP secret doesn't match this client code: re-enable TOTP and save the new secret"), available cash, holdings count and value. Never returns tokens; masks the client code; rate-limited to 1 call per 2 minutes (avoid lockouts)	fake-SmartConnect tests for OK, AB1050, network error; a second call within 2 min gets 429
V0-7	Upgrade `smartapi-python` from the pinned **1.4.1** (old base URL `apiconnect.angelbroking.com`) to **1.5.5** (Feb 2025, base URL `apiconnect.angelone.in` , matches Angel's current docs). `generateSession(client_code, PIN, pyotp.TOTP(secret).now())` per the official README. Re-record the login fixture	fake-server login test passes on 1.5.5; `scripts/login_check.py` prints 4 OK lines on the server (owner runs)
V0-8	**Keys page: see what is saved** (owner, 23 Oct 2026: "give option to see old values on keys page"). (a) Every row always shows a **masked preview** (first 2 + `•••` + last 2; MPIN shows `••••` only), length, saved time, source (DB / .env) and a 12-char **keyed** fingerprint (HMAC-SHA256 with `SECRETS_MASTER_KEY` , so it can't be reversed by guessing). **No fingerprint for the MPIN or any value under 8 characters**: a plain hash of a 4-digit PIN is reversed by trying all 10,000 PINs (lesson from 23 Oct 2026); check those with Compare only. (b) **Compare**: type a value → `POST /api/keys/compare {key, value}` → "matches the saved value" / "differs" (+ which position first differs, e.g. "character 5"); nothing is returned but the verdict. (c) **Reveal** (eye button): `POST /api/keys/reveal {key, password}` needs the admin cookie **and** the admin password typed again; returns the current value once with `Cache-Control: no-store` ; the page shows it for 30 s with a copy button, then hides it; writes event `key_revealed` (key name only); max 5 reveals per 10 min. (d) **History**: new table `app_settings_history(key, fingerprint, masked, saved_at, saved_by)` (keyed fingerprint; none for short values) appended on every save/clear; the page lists the last 5 versions per key, masked + fingerprint + time. Old plaintext is never kept (a replaced secret is gone by design). Compare and reveal only work over HTTPS, Tailscale or localhost (same rule as save, V0-2); on plain HTTP they're greyed with "needs a secure connection"	tests: masked preview for each kind; compare true/false + first differing position; reveal 401 without cookie, 403 with a wrong password, 403 over plain HTTP from the internet, 429 on the 6th reveal; history row on every save; no value in any log line or event
V0-5	B23: one path source. Every script defaults `--db` / `--lake` from `settings.data_dir` . Move the lake to `var/lake/` (keep `data/` as the Python package only)	test: research and portal resolve the same db path

V1: MARKET LAYER (`FEAT/V1-MARKET`)

TICKET	WORK	DONE WHEN
V1-1	B16: `config/sessions.json` with dated regimes (MARKET_INDIA §2): `{from, to, class: "fo"|"non_fo", pre_open: [...phases], continuous: ["09:15", end], cas: {...}|null, post_close: ["15:50","16:00"]}` . `session_state(ms, stock_class)` returns PRE_OPEN, OPEN, NO_NEW_ENTRIES, SQUARE_OFF, CLOSING_AUCTION, POST_CLOSE, CLOSED, HOLIDAY, SPECIAL	tests: F&O stock 23 Oct 2026 15:20 → CLOSING_AUCTION; same time 8 Dec 2026 → OPEN; non-F&O 23 Oct 15:20 → OPEN; 09:09 → PRE_OPEN; Muhurat 8 Nov 14:30 → SPECIAL/OPEN
V1-2	B17/B18: Muhurat 2026 per NSE (afternoon; owner verifies Q4). Calendar loads every `holidays_*.json` ; asking about a year without a file raises `CalendarMissing` ; health turns red; all trading jobs skip with an alert. Every cron job checks `is_trading_day` first. Muhurat: no new entries in either book (setting `trade_special_sessions=false`)	tests: 2027-01-04 raises without a 2027 file; a 2026 holiday skips jobs; Muhurat window correct
V1-3	B19/D4: derive `square_off_at` / `no_entry_after` from `broker_squareoff_at` and the leads. Book A shadow uses the same times	tests: broker 15:10 → engine 14:55 / 14:35; invariant test
V1-4	B20 + D3: refresh `costs_angel.json` (MARKET_INDIA §5) with `source_url` , `as_of` , `verified_against_contract_note: false` ; `order_brokerage` = max(min ₹5, min(₹20, 0.1%)); risk enforces `min_position_cnc_inr` with reason `below_min_cnc_position`	tests: the round-trip table in MARKET_INDIA §5 to the paisa; ₹3,000 order → ₹5 brokerage
V1-5	Instruments: daily scrip master → `tick_size` , `is_fo` (symbol has NFO futures), series EQ only; `config/surveillance.json` (ASM/GSM) with risk reason `surveillance` ; B26 exposure includes the new position	fixture tests; a surveillance-listed symbol is skipped with the reason
V1-6	Survivorship: `config/universe_history.csv` (date, symbol, action add/remove) + `universe_on(date)` ; lab uses it	test: a symbol removed in 2020 is tradable in 2019 and not in 2021

V1-7	B21: UDiFF bhavcopy parser (check exact headers against one real file the owner downloads; commit a 5-row fixture), legacy parser for old dates, delivery file, `data/corporate_actions.py` (split/bonus factors from NSE corporate actions; the owner provides an export for the fixture). The lake stores raw OHLC plus `adj_factor`; strategies read adjusted prices; costs use raw prices	tests: both bhav formats; a 1:1 bonus fixture gives no −50% gap in adjusted closes
V1-8	`data/events.py`: results dates (NSE board-meeting / financial-results announcements), F&O expiry = last Tuesday of the month (moved back if a holiday), RBI MPC dates (config), Budget day. Features on every signal: `results_in_days`, `expiry_day`, `mpc_day`	tests: expiry-day rule on 2026 months incl. a holiday Tuesday

V1B: OPENING AND SENTIMENT RECORDER (`FEAT/V1B-OPENING`), D8

Daily timeline (IST): 06:00–23:30 global cues every 5 min · 08:30 overnight headlines scored · **08:45 morning brief frozen** (Telegram) · 09:00–09:15 pre-open snapshots every 30 s · 09:15–09:45 opening scorecard every minute · 09:20 Telegram opening summary · 15:45 opening type and open-to-close written · 18:00–20:00 FII/DII provisional · every 15 min (08:00–23:30) headlines fetched, tagged and scored.

TICKET	WORK	DONE WHEN
V1b-1	`data/global_cues.py`: instrument map in `config/cues.json` (key, label, region, source symbol: Yahoo `^GSPC`, `^IXIC`, `^DJI`, `ES=F`, `NQ=F`, `^GDAXI`, `^FTSE`, `^FCHI`, `^STOXX50E`, `^N225`, `^HSI`, `^KS11`, `INR=X`, `DX-Y.NYB`, `BZ=F`, `GC=F`, `^TNX`, `^NSEI`, `^INDIAVIX`; GIFT Nifty via `gift_nifty_source`). Each row stores its own previous close and a `stale` flag (market closed, or no update for 2× its interval). `data/flows.py`: NSE FII/DII provisional (browser-like session; unofficial)	fixture tests per source; a closed market is marked stale, not 0%
V1b-2	`data/preopen.py`: NSE pre-open JSON (`key=NIFTY` and `key=FO`) every `preopen_snapshot_s` from 09:00 to 09:15; the final snapshot is marked `final=1`; the Nifty IEP too. If NSE blocks the request, try Angel's quote during pre-open (VERIFY), else log `preopen_unavailable`	fixture test; a missing day shows on Data health
V1b-3	News: real scheduled fetch (Moneycontrol, ET Markets, Business Standard, Reuters India, NSE announcements RSS where available), `published_ms` from the feed, dedupe by `external_id` and near-duplicate title; new tags (MARKET_INDIA §8b); rules-only scoring on the server (FinBERT via the HF Space optional, same as crypto); `headlines` table; each headline later gets `nifty_move_next_60m`	tag tests for each new event type; dedupe test
V1b-4	`engine/opening.py`: build and freeze the morning brief at 08:45; opening scorecard; opening type at 15:45 (rules: gap ≥ +0.5% & close > open = gap-up-and-go; gap ≥ +0.5% & close < open = gap-up-fade; mirror for gap-down; else flat); `sentiment_daily` composite (reuse `news/sentiment.composite_score`, parts stored separately); risk-off alarm rule + one Telegram alert per day	tests: each opening type on a hand-built day; alarm fires only when all three conditions hold within the window
V1b-5	Endpoints `/api/opening`, `/api/sentiment`, `/api/headlines` (§7) and the Market mood page (§8) (if V5 isn't merged yet: JSON + a temporary page)	contract tests; owner sees the morning brief on the phone at 08:45
V1b-6	Research hook: every Book A/B signal gets the day's `gap_pct`, `imbalance`, `composite`, `open_type_so_far` and `risk_off_alarm` as features (no lookahead: only values known at decision time)	test: a 10:00 signal never sees the 15:45 opening type

V2: BACKTESTER CORRECTNESS (`FEAT/V2-LAB-FIXES`)

Write each bug as a failing test first. The probes in §3.3 are the test cases.

TICKET	WORK	DONE WHEN
V2-1	B3/B4: daily fills. Entry is the next day's LIMIT at close × 1.005, filled only if that day's low ≤ limit, at min(open, limit) + slippage; else `no_fill`. Stop: if open ≤ stop, fill at open − slippage, else at stop − slippage (− extra 0.05% for stops, as PLAN A9.5). Trail = highest close since entry − 2.5×ATR14, never lowered. Time exit after 20 sessions at the close. Entry day: the stop applies from the open (conservative). `entry_ms` = fill day 09:15 IST	tests: gap-down probe → R ≈ −10.6; gap-up over the limit → no_fill; trail never lowers
V2-2	B1/B2/B14/B15: intraday. Resample 1m → strategy timeframe before calling the strategy. Build the context every day (`prev_close`, `pdh`, `gap_pct`, `slot_vol_avg` from 20 prior days, cross-sectional `ret_decile` / `rs_rank` across the universe at the decision minute). Stops for both sides; targets; time exit at the engine square-off; one trade per strategy, symbol and day	tests: the short probe stops at 103.2 ± slippage; ORB target 2R hit; square-off at 14:55 (V1-3 times); S3/S4/S5 each fire on a hand-built day
V2-3	B13: BarView knows the timeframe. A bar is closed at open + timeframe; a daily bar closes at the session's close for its class and date (V1-1). Asking for an unclosed bar raises `LookaheadError`	tests: daily bar not visible at 14:00 the same day; visible at 16:00
V2-4	B5: `lab/baselines.py` simulates real mirror trades (opposite side, same stop % and target R, same exit engine) and random-time trades (deterministic seed, same day/symbol window, same side, stop % and exits; daily: a random day within ±10 sessions). Delete the fabricated baselines in `scripts/research.py`	test: on a strategy with planted edge, pick > mirror and > random; on shuffled outcomes the gate fails
V2-5	B6: portfolio backtest over `universe_on(date)` with shared cash, `max_open_swing`, one position per symbol, priority when several signals fire the same day (crypto `swing_book.priority`: strongest strategy first, then lower ATR%), costs per trade	tests: two simultaneous signals with 1 free slot → the priority one taken, the other logged `max_open_swing`; deterministic
V2-6	Evaluate: halves by date, by year, by India VIX tercile, by results season, cost stress ×1.5; ledger stores `prereg_id`, `data_hash`, `post_hoc` columns	tests: same inputs → same report hash; trials count increments
V2-7	D6: `core/exits.py` with `fill_entry(order, bar)`, `step_exits(position, bar, rules) → (position, exit|None)`, `virtual_outcome(signal, bars, rules)`. `lab/backtest.py`, the paper engine (V4) and the outcome job (V4-4) all call it	test: backtest trade == paper trade == virtual outcome for the same signal and bars

V3: REAL DATA AND THE RESEARCH RUN (`FEAT/V3-RESEARCH`)

TICKET	WORK	DONE WHEN
V3-1	Backfill real bhavcopy 2012-01 → today (owner runs the downloader; throttled, resumable), apply corporate actions, data-health report. Delete the synthetic lake	`docs/phase-reports/V3.md` lists symbols, first/last date, missing days (0 on trading days), corporate actions applied
V3-2	Write `docs/PREREG.md` **before** running: S6_swing, S6b_keltner (PLAN A10 as written, now with correct fills); F1 = S6 + Nifty 30-day vol rank > 0.667 skip (crypto filter port); F2 = S6 + skip when Nifty < 200-day SMA; S7_xsmom (first trading day of each month: rank `universe_on(date)` by the mean of z-scores of 6m and 12m return ÷ daily-return volatility, like Nifty200 Momentum 30; hold the top N = floor(capital / min_position_cnc_inr); exit when a holding falls out of the top 2N at a rebalance or at a 3×ATR20 safety stop). 5 trials total, halves 2012–2018 / 2019–2026	PREREG committed in a commit **before** the run commit
V3-3	Run all five through the lab (V2), Book B gate (n ≥ 100 per half, avg R ≥ +0.10 both halves, beats mirror and random both halves, R > 0 under cost stress ×1.5). Book A only on whatever 1m depth exists (likely none)	`V3.md` with the gate table, by year, VIX tercile, results season, cost in R at ₹1L capital, and a plain-English verdict per trial; ledger count shown

The owner then picks which passed trials (if any) go to paper. "None passed" is a valid outcome. Then the paper engine still runs S6 **as a shadow book** to collect live evidence (same as crypto kept logging).

V4: PAPER ENGINE, BOOK B END-OF-DAY (`FEAT/V4-PAPER`)

Daily timeline (trading days only, IST, times from `sessions.json`):

TIME	JOB	WHAT IT DOES
08:30	`job_instruments`	scrip master, ticks, F&O flags, surveillance list, today's session regime
08:45	`job_login`	Angel login (data only while paper); on failure the paper engine still runs on bhavcopy

09:00	`job_pending`	move `pending_orders` for today into the PaperBroker as DAY LIMIT orders
09:15–15:30	`job_mark` every 5 min	Angel LTP (Yahoo fallback) → unrealised P&L for display only. **Never used for fills**
16:15	`job_eod_fill`	today's daily bar per symbol (Angel ONE_DAY after CAS + post-close): fill pending entries by `core/exits.fill_entry`; step every open position by `step_exits`; book costs; close trades; update the cycle wallet
16:20	`job_scan`	run enabled Book B strategies on today's closed daily bars; risk; write signals (pick + mirror + random); for picks that pass, write `pending_orders` for the next trading day
16:30	`job_outcomes`	`virtual_outcome` for every unresolved signal (taken, skipped, mirror, random)
16:35	`job_eod_report`	`daily_pnl`, Telegram summary
18:30	`job_bhav_check`	bhavcopy arrives; re-check today's bars; any difference to the Angel bar > 1 tick → parity event (bhavcopy is truth for research)
18:45	`job_backup`	as PLAN A14.2

TICKET	WORK	DONE WHEN
V4-1	Migration 002 (§5); repository functions	migrate twice is a no-op; insert/read each new table
V4-2	`data/quotes.py`: Angel LTP poller (batched, throttled) with Yahoo fallback; writes the quotes cache used by the tape and `job_mark`	fake-client tests; the fallback is used when Angel is unavailable
V4-3	B7: PaperBroker accounting. `cash` += realised P&L − costs; CNC buy needs cash ≥ value + estimated costs (else REJECTED `insufficient_funds`); CNC sell only from holdings (no short); holdings bought today can't be sold today; MIS margin = 20% of value, released with P&L. Equity = cash + Σ holdings × mark	tests: the +₹10k probe ends at ₹1,10,000 − costs; CNC short rejected; same-day CNC sell rejected
V4-4	`engine/outcomes.py` + `job_outcomes`; take vs skip scoreboard (crypto `regime_gate.signal_scoreboard`)	tests: skipped signal resolves to the same R the trade would have had
V4-5	B8/B10/B11/B12: Broker protocol (sync, dataclasses) implemented by PaperBroker and an Angel adapter; OrderManager looks up the tag before sending (real idempotency); watchdog tracks `crossed_since_ms` per bracket and fires only after `stop_watchdog_s` continuous crossing; `square_off` sends a protected LIMIT exit	tests per PLAN A15 orders list, incl. the watchdog probe now **not** firing on the first tick
V4-6	`engine/paper_engine.py` + runner wiring for the timeline above; cycle start on first enable; cycle ends on target / floor; kill switch stops new entries only	an end-to-end test replays 30 fixture days through the jobs and gets the same trades as `lab/backtest.py` on those bars (parity = 0 differences)
V4-7	Telegram: entry, exit, rejection, daily summary (PLAN A13.3 wording), quiet hours	fake Telegram tests
V4-8	Parity job (PLAN A12.2) on the paper book: replay today vs paper trades	parity page shows 0 faults on the e2e fixture

V5: PORTAL PARITY WITH CRYPTO (`FEAT/V5-PORTAL`, CAN START AFTER V0)

TICKET	WORK	DONE WHEN
V5-1	Copy crypto `scheduler/portal/{core.js, portal.css, portal.html, nav.js, chrome.js, chrome.css}` into `web/portal/`; adapt to hash routing under `<base href="/nse/">`, the NSE nav (§8) and versioned assets (`?v=<hash>`, long cache)	the shell loads at `/nse/`; four palettes switch and persist
V5-2	Endpoints in §7 (`web/portal_api.py`), backed by real tables when present and **empty states** when not; `web/reasons.py`	contract tests: each endpoint matches its fixture's keys
V5-3	Command hub: Today and Market mood (§8)	widget/view/store structural test passes
V5-4	Book hub: Paper book and Signals	same, plus a test that no widget renders demo numbers when the store is empty
V5-5	Evidence, Health and Settings (Go-live gates computed server-side in `/api/golive`)	same
V5-6	Phone check: each page at 375 px wide, no horizontal scroll; `prefers-reduced-motion` honoured	owner checks on the phone (PLAN A1 working style)

V6: BOOK A SHADOW ON THE LIVE FEED (`FEAT/V6-INTRADAY-SHADOW`)

TICKET	WORK	DONE WHEN
V6-1	`data/stream.py`: real Angel WebSocket V2 thread, SnapQuote, reconnect/backoff, resubscribe, ticks → asyncio queue (PLAN P1-5)	fake-WS reconnect test
V6-2	`data/bars.py`: implement the late-tick rule (it's a `pass` today), F&O stocks' last bar 15:14 while CAS runs; record `bars_1m` and `spread_1m`	late / out-of-order / missing-minute tests
V6-3	Book A strategies on live bars → signals with role and decision `shadow` → `virtual_outcome` with MIS costs at the engine square-off. No paper trades while `book_a_mode=shadow`	a recorded fixture day produces the same shadow signals as the backtest on the same bars
V6-4	Angel 1m history depth report (PLAN P1-3). The paid-feed decision stays with the owner	report per symbol

V7: LIVE READINESS (FORMER P5; `FEAT/V7-LIVE`), THIS PHASE CAN SEND REAL ORDERS

As PLAN A16-P5, plus: Angel adapter sends LIMIT / STOPLOSS_LIMIT only (no MARKET, no IOC: Angel prohibits them for algos); static-IP check before the first order of the day; the IP can change at most once a week, so moving servers is planned ahead; CAS-aware cut-offs; contract-note reconciliation of costs (paper model within +0.03%); CNC stop orders re-placed every morning (DAY orders expire); tax export by category (MARKET_INDIA §7). Nothing merges without the owner's written yes.

10. Prompts to paste into Cursor (one chat per phase)

```
[V0] Read AGENTS.md, docs/PLAN_V2.md (§3, §6, §9-V0) and docs/MARKET_INDIA.md §4. Branch feat/v0-hygiene from
feat/mvp-backtest-news after the owner has committed it. Implement V0-1..V0-5 exactly. Write the failing test
first for each ticket. Never change mode/live_orders defaults. ruff + pytest green, docs/phase-reports/V0.md, PR.

[V1] Read AGENTS.md, docs/PLAN_V2.md (§3.3 B16-B21, B26, §4, §6, §9-V1) and docs/MARKET_INDIA.md fully. Branch
feat/v1-market. Implement V1-1..V1-8. Session times, holidays, ticks and costs are data files with dates and
sources, never constants in code. Use the test cases written in each ticket. Where MARKET_INDIA says VERIFY,
keep the value configurable and list it in V1.md under "owner to verify". PR.

[V1b] Read AGENTS.md, docs/PLAN_V2.md (§2 D8, §5 opening tables, §6, §7 opening/sentiment/headlines, §8
Opening + Sentiment pages, §9-V1b) and docs/MARKET_INDIA.md §8b. Branch feat/v1b-opening. Build the recorder:
global cues, FII/DII, pre-open snapshots, scheduled tagged headlines, morning brief, opening scorecard, opening
type, sentiment composite, risk-off alarm. It is recording + display only: nothing here may block, size or
time a trade. Every source has a fixture test and a stale flag. No network in tests. PR.

[V2] Read AGENTS.md, docs/PLAN_V2.md (§3.3 B1-B6, B13-B15, §9-V2) and docs/PLAN.md A10-A11. Branch
feat/v2-lab-fixes. First add the probe cases from §3.3 as failing tests, then fix. Build core/exits.py and make
lab/backtest.py use it. Do not change any strategy parameter (pre-registered); if a rule is ambiguous, ask.
Delete the fabricated baselines. PR with before/after numbers on the synthetic fixture only.

[V3] Read AGENTS.md, docs/PLAN_V2.md (§9-V3) and docs/MARKET_INDIA.md §9-10. Branch feat/v3-research. Write
docs/PREREG.md and commit it before any run. I (owner) run the bhavcopy backfill on my machine with the command
you give me. Run the five trials, count every run in the ledger, write docs/phase-reports/V3.md with the gate
table and a plain-English verdict. No parameter changes after seeing results (a change = new post-hoc trial).

[V4] Read AGENTS.md, docs/PLAN_V2.md (§2, §5, §7, §9-V4) and docs/PLAN.md A9, A12. Branch feat/v4-paper. Build
the Book B end-of-day paper engine with the timeline in V4. Paper must equal the backtest on the same bars
(V4-6 e2e test). Fix PaperBroker accounting first (B7). No live orders anywhere. PR.

[V5] Read AGENTS.md, docs/PLAN_V2.md (§7, §8, §9-V5) and the crypto repo's docs/PORTAL_REVAMP.md plus
scheduler/portal/ at /Users/dhruv/personal/Tennis-bet/crypto-signal-engine. Branch feat/v5-portal. Port the
micro-UI and build every page in §8 against the §7 endpoints (fixtures first). Nothing simulated on the real
portal; empty states instead. Port crypto's structural portal tests. PR with phone screenshots.

[V6] Read AGENTS.md, docs/PLAN_V2.md (§9-V6), docs/PLAN.md A8, P1-5..P1-7. Branch feat/v6-intraday-shadow.
Real Angel WebSocket, bar recording, Book A in shadow only. PR.

[V7] Read AGENTS.md, docs/PLAN.md A3, A8, A9.4, A12.3, A14, A16-P5 and docs/PLAN_V2.md §9-V7. Branch
feat/v7-live. THIS PHASE CAN SEND REAL ORDERS: put the warning in capitals at the top of the PR. Implement
behind mode=live AND live_orders=true. Do not set either flag anywhere.
```

11. Questions for the owner (Cursor uses the default until answered)

#	QUESTION	DEFAULT
Q1	What is Angel's MIS auto square-off time for F&O stocks right now? (Angel app → order screen, or support)	15:10 → engine square-off 14:55, no entries after 14:35
Q2	Portal access: keep public `http://52.62.37.4/nse/`, or move to Tailscale (PLAN A13) / add HTTPS?	Tailscale; `/keys` refuses plain HTTP from the internet
Q3	Book A shadow-only until it passes the gate (like crypto intraday)?	Yes
Q4	Verify Muhurat 2026 (8 Nov) timings in the NSE circular. Should either book trade it?	14:20–15:15 continuous; no new entries in either book
Q5	Paper capital ₹1,00,000 and min CNC position ₹20,000?	Yes
Q6	Approve the 5 pre-registered Book B trials (S6, S6b, F1, F2, S7)?	Yes
Q7	Paper cycle target wallet (crypto uses a target)?	None, open-ended
Q8	Commit the current untracked tree to `feat/mvp-backtest-news` and open the MVP PR?	Yes
Q9	Share the first contract note (masked) so costs can be marked verified?	Pending
Q10	Download one UDiFF bhavcopy and one NSE corporate-actions export for fixtures?	Pending
Q11	GIFT Nifty data: no reliable free API. Pay for a feed, or use US futures + Asia as the proxy?	Proxy until decided
Q12	OK to read NSE website JSON (pre-open, FII/DII) with a browser-like session? It's unofficial and can break	Yes, with a stale flag and alerts
Q13	Angel login fails with AB1050 "Invalid totp and client combination" (23 Oct 17:10 IST). Please re-enable TOTP on smartapi.angelone.in for this client code, save the new secret on `/keys`, then rerun the read-only check	Blocking anything that needs Angel

12. Risks specific to this plan

1. **Session rules keep changing** (CAS on 3 Aug, revised for 7 Dec 2026, unfilled-order rules on 8 Feb 2027). Mitigation: dated `sessions.json`, a health check that fails when today falls outside every regime.
2. **Corporate-action data quality.** A missed bonus fakes a crash. Mitigation: a gap > 15% on a Nifty 100 stock with no action recorded raises a data-health alert and excludes that day from research.
3. **Small Book B sample.** Daily breakouts on 100 stocks give maybe 100–300 trades per half. Mitigation: the n ≥ 100 rule, plus "no" is an accepted answer.
4. **Momentum crashes** (−30% in Sep 2024–Mar 2025 for the momentum index). Mitigation: report the worst 3-month window and the worst year in V3, not just the average.
5. **Paper EOD fills are optimistic about queue position** at the limit price. Mitigation: a fill needs the day's low to be at least one tick **below** the limit (not just equal), and slippage applies.
6. **Copying the crypto UI brings its complexity.** Mitigation: port only the pages in §8; no AI pages until news or AI proves lift.

Appendix B · Full market research (docs/MARKET_INDIA.md)

Written 23 Oct 2026 by Claude (planner) for Cursor (builder) and the owner. This is the market knowledge behind `docs/PLAN_V2.md`. Where it disagrees with `docs/PLAN.md`, this file is newer. Facts marked **VERIFY** had conflicting sources on 23 Oct 2026; the owner confirms them (contract note, Angel app, NSE circular) before paper or live depends on them. Sources are listed at the end.

1. The pieces

PIECE	WHAT IT IS	WHY THE ENGINE CARES
SEBI	The market regulator	Sets the algo rules (§6), margin rules (§3), closing auction (§2)
NSE	The exchange we trade on (cash market, "CM" segment)	Sessions, ticks, price bands, bhavcopy, holidays
NSE Clearing	Settles trades	T+1 settlement (§3)
CDSL	Depository that holds Angel clients' shares	DP charge on every delivery sell day (§5)
Angel One	Our broker; SmartAPI is its API	Costs, auto square-off time, static IP, order rules
Nifty 50 / Nifty Next 50	Two indices; together they make the **Nifty 100** (our universe)	Rebalanced twice a year (March and September), so old data needs past members (survivorship, §9)
India VIX	Volatility index built from Nifty options	Market-mood feature; candidate market filter
GIFT Nifty	Nifty futures on NSE IX in GIFT City, trading almost round the clock (replaced SGX Nifty in July 2023)	Best overnight hint of where Nifty will open

The cash market is a continuous limit order book: orders match on price, then time. Each stock trades in a **series**. `EQ` is normal rolling settlement (intraday allowed). `BE` / `BZ` are trade-for-trade (every trade must be delivered, no intraday). The engine trades `EQ` only. All Nifty 100 stocks are `EQ`.

Most Nifty 100 stocks also have futures and options. That makes them **"F&O stocks"**, and since August 2026 they follow different closing rules from other stocks (§2). The engine must know, per symbol, whether it is an F&O stock.

2. The trading day (IST)

PRE-OPEN CALL AUCTION, 09:00–09:15 (CHANGED 7 SEP 2026)

TIME	PHASE
09:00–09:05	Order entry: limit and market orders
09:05–09:10	Limit orders only. Market orders can't be changed or cancelled. Entry closes **at a random moment between 09:08 and 09:10**
09:10–09:12	Matching at one equilibrium price, which becomes the day's open
09:12–09:15	Unmatched orders move into continuous trading

CONTINUOUS TRADING AND THE CLOSE: IT DEPENDS ON THE DATE AND ON THE STOCK

The closing auction session (**CAS**) replaced "close = VWAP of the last 30 minutes" for F&O stocks. Its timings changed once already:

PERIOD	F&O STOCKS (MOST OF THE NIFTY 100)	NON-F&O STOCKS
Before 3 Aug 2026	Continuous 09:15–15:30. Close = VWAP of 15:00–15:30	Same
3 Aug 2026 – 6 Dec 2026 (now)	Continuous **09:15–15:15**. CAS 15:15–15:35: 15:15–15:20 reference price + transition, 15:20–15:25 limit + market entry, 15:25–15:30 limit-only entry (random close 15:28–15:30), 15:30–15:35 matching. **Close = auction price**	Continuous 09:15–15:30. Close = 30-min VWAP
From 7 Dec 2026 (SEBI circular 15 Oct 2026)	Continuous **09:15–15:30**. CAS 15:30–15:40: 15:30 reference price (1 min), 15:31–15:36 limit + market entry, 15:36–15:40 limit-only entry (random close 15:38–15:40). Rules for orders left unfilled change again on 8 Feb 2027	Unchanged (**VERIFY** that non-F&O stocks still close on VWAP)
All	Post-close session 15:50–16:00, trades at the closing price only	Same

What this means for the engine:

- **Session times are data, not constants.** They need an effective-from date and a stock class (F&O or not). `core/clock.py` today assumes a 15:30 close for every stock on every date.
- **Daily bars changed meaning.** For F&O stocks, the daily "close" since 3 Aug 2026 is an auction price. From 7 Dec it is also set after 15:30. A backtest should flag this regime change (a feature column) and not treat the close as a tradable continuous-market price.
- **No 1-minute bars exist after 15:15** for F&O stocks between 3 Aug and 6 Dec 2026. Bar-health checks must not count those minutes as missing.
- The engine never trades in the pre-open or the CAS. Book B's next-day entries are placed at or after 09:15.

BROKER AUTO SQUARE-OFF (MIS), VERIFY

Angel One's support page says intraday equity square-off is at **15:15**, but that page predates CAS. Since 3 Aug 2026 brokers moved the cut-off earlier for F&O stocks, to between 15:00 and 15:10 (Fyers 15:05, 5paisa 15:10, JM 15:05, share.market 15:00). One comparison site reports Angel at 15:10 or 15:15. Angel charges its normal brokerage for auto square-off, with no extra penalty. **The engine's 15:05 square-off (PLAN A3) can now land after the broker's.** The owner checks the cut-off in the Angel app. The engine then squares off at least 15 minutes before it (PLAN_V2 V1-3).

SPECIAL SESSIONS 2026

- **Muhurat trading: Sunday 8 Nov 2026, afternoon, not evening.** NSE's notice (reported 19 Oct 2026) gives a block-deal window 13:50–14:05, pre-open 14:05–14:15, and continuous trading 14:20–15:15, then CAS. Sources disagree on the exact end (15:15 + CAS vs 15:30), so **VERIFY** against the NSE circular. `config/holidays_2026.json` has a placeholder of 18:15–19:15, which is wrong. Recommendation: no new entries in either book during Muhurat (one thin hour). Book A stays flat.
- The 2026 Union Budget (Sunday 1 Feb) had a special live session. Such days are announced by circular and belong in the special-sessions list.

HOLIDAYS 2026 (WEEKDAYS THE MARKET IS SHUT)

26 Jan, 3 Mar, 26 Mar, 31 Mar, 3 Apr, 14 Apr, 1 May, 28 May, 26 Jun, 14 Sep, 2 Oct, 20 Oct, 10 Nov, 24 Nov, 25 Dec. Some lists add 15 Jan (Maharashtra civic elections). Weekend festivals cause no extra closures. NSE publishes next year's list around December. **The engine must refuse to trade a year it has no holiday file for** instead of treating every weekday as open, which is what it does today.

3. Products, margin and settlement

CNC (DELIVERY), USED BY BOOK B	MIS (INTRADAY), USED BY BOOK A

Direction	Long only. Selling shares you don't hold creates a short delivery, which goes to auction with a penalty	Long or short
Holding	Overnight, as long as you like	Must be closed the same day, or the broker squares it off
Money needed	Full value	Upfront margin of at least VaR + ELM, with a 20% floor, so at most about **5x** leverage. Brokers can't promise fixed leverage. Peak-margin snapshots during the day enforce it
Settlement	**T+1**: shares reach the demat account the next trading day. Same-day sale proceeds can usually fund new buys	Netted the same day
Costs	STT 0.1% on both sides, stamp 0.015% on the buy, DP charge on the sell day	STT 0.025% on the sell, stamp 0.003% on the buy, no DP
Tax	Capital gains (§7)	Speculative business income (§7)

Buying CNC and selling the same day counts as an intraday trade for STT and tax. Book B must hold for at least one night. T+0 settlement (optional, limited list) and MTF (margin funding) are out of scope.

4. Prices: ticks, bands and surveillance

Tick size by price band (NSE, from 15 Apr 2025, reviewed monthly on month-end prices):

PRICE	TICK
below ₹250	₹0.01
₹250 – ₹1,000	₹0.05
₹1,000 – ₹5,000	₹0.10
₹5,000 – ₹10,000	₹0.50
₹10,000 – ₹20,000	₹1.00
above ₹20,000	₹5.00

A stock's tick can change at a month boundary, so **read `tick_size` from Angel's scrip master every morning**. This table is only the fallback, and the engine must never assume ₹0.05. An order price off the tick is rejected.

Price bands. Non-F&O stocks have fixed daily bands (2/5/10/20%). F&O stocks have **dynamic** bands: an operating range (typically 10%) that the exchange widens step by step when price presses against it. They have no permanent "upper circuit", but you still can't trade outside the current range. The full quote's upper/lower limits are what the circuit-buffer check must use. Market-wide circuit breakers (10/15/20% moves in Nifty or Sensex) halt everything.

Surveillance lists. ASM (Additional Surveillance Measure) and GSM (Graded Surveillance Measure) put stocks on higher margins, trade-for-trade or tighter bands. NSE publishes the lists daily. Nifty 100 stocks rarely appear, but the engine should read the list and exclude anything on it (skip reason `surveillance`).

5. What a trade costs at Angel One (Oct 2026)

CHARGE	DELIVERY (CNC)	INTRADAY (MIS)	WHERE IT'S FROM
Brokerage	lower of ₹20 or 0.1% per executed order, **minimum ₹5**	same	Angel pricing update, effective 17 Nov 2025
STT	0.1% buy + 0.1% sell	0.025% sell	Unchanged by Budget 2026, which only raised F&O STT
NSE transaction charge	**0.00307%** each side (Angel lists 0.0030699%)	same	Angel and Zerodha tables. The code has 0.00297% (the Oct 2024 rate). **VERIFY** on a contract note
SEBI fee	₹10 per crore each side	same	
Stamp duty	0.015% of buy	0.003% of buy	
GST	18% of (brokerage + transaction charge + SEBI fee), and on DP	same	
DP charge	₹20 + GST = **₹23.60 per stock per sell day**	none	Angel support page
IPFT	0.0000001%, too small to model	same	
Auto square-off	n/a	charged like normal brokerage (no penalty)	Angel charges page

Some marketing pages still say "₹0 delivery brokerage". Angel's own pricing page says ₹20 / 0.1% with a ₹5 minimum. The first real contract note settles it.

Round trip, recomputed with these rates (buy and sell the same value):

BOOK	POSITION	CHARGES	% OF POSITION	+ SLIPPAGE	IN R
MIS	₹25,000	₹56.07	0.224%	0.284% (0.06%)	0.57R at a 0.5% stop, 0.28R at 1%
MIS	₹50,000	₹64.94	0.130%	0.190%	0.38R / 0.19R
MIS	₹1,00,000	₹82.68	0.083%	0.143%	0.29R / 0.14R
MIS	₹2,00,000	₹118.16	0.059%	0.119%	0.24R / 0.12R
CNC	₹10,000	₹69.45	0.694%	0.794% (0.10%)	0.16R at a 5% stop, 0.10R at 8%
CNC	₹25,000	₹126.42	0.506%	0.606%	0.12R / 0.08R
CNC	₹50,000	₹182.04	0.364%	0.464%	0.09R / 0.06R

Two consequences that PLAN.md missed:

1. **Book B slot size matters.** PLAN sizes Book B as capital ÷ `max_open_swing`. At ₹50k and 5 slots that is ₹10k a position, which costs 0.79% a round trip (the flat ₹23.60 DP charge and the ₹20 brokerage dominate). At ₹25k it is ₹5k positions and about 1%. Keep CNC positions at **₹20k or more**: fewer slots, more capital, or both (PLAN_V2 V1-4).
2. **Book A can't trade at ₹50k under its own rules.** At 1% risk (₹500) and the ₹1L minimum position, the stop must be 0.5% or tighter, which is exactly where costs take 0.29R a trade. The 0.8–1.5% stops PLAN asks for need ₹800–1,500 risk, which is 1.6–3% of ₹50k. Book A stays shadow-only until capital and evidence change that (PLAN_V2 §2).

6. Rules for API (algo) trading, Angel SmartAPI, from 1 Apr 2026

- **Static IP:** API orders are accepted only from the registered primary static IP. A secondary IP can be added. The mapped IP can be changed **at most once a calendar week**. Data and login calls are not tied to the IP.

- **Order types: market and IOC orders are prohibited** for algo orders (some brokers convert market orders to market-price-protection orders, but Angel prohibits them). The engine uses LIMIT and STOPLOSS_LIMIT only, which PLAN already says.
- **Speed:** a 10 orders-per-second threshold per exchange segment, counting new orders, modifications and cancels. Below it no strategy registration is needed and orders carry a generic algo ID. We throttle at 5/s.
- **Login:** one session per trading day with 2FA (client code + MPIN + TOTP). No long-lived refresh tokens.
- **Hosting:** retail algos run on the broker's servers, except "tech-savvy" clients who run their own logic from a registered static IP. That is us (EC2 Elastic IP).

7. Tax (for the CA export)

- **Delivery (Book B):** short-term capital gains (held 12 months or less) are taxed at **20%**. Long-term gains are taxed at **12.5%** above ₹1.25 lakh a year. Budget 2026 didn't change these.
- **Intraday (Book A):** speculative business income, taxed at the slab rate, filed in an ITR-3 style return.
- The **Income Tax Act 2025** replaces the 1961 Act from tax year 2026-27, and section numbers changed. The tax export should label rows by category (intraday-speculative / STCG / LTCG / charges split), not by section number. The owner's CA maps them.

8. What moves Indian large caps (feature ideas, not proven edges)

DRIVER	WHEN	NOTES FOR FEATURES
Global cues	Overnight	US close, GIFT Nifty change since 15:30 IST, Asian open
FII / DII cash flows	NSE publishes provisional numbers every evening	Foreign selling has driven most big Nifty drawdowns. A flow is known only after the close, so it's a next-day feature (no lookahead)
USD/INR, crude oil	Continuous	India imports most of its oil, so crude spikes hurt OMCs, paints, aviation and the rupee
RBI policy (MPC)	6 meetings a year, decision about 10:00 IST	Banks and NBFCs react
Union Budget	1 Feb	Big sector moves. Special session if it falls on a weekend
Quarterly results	Jan, Apr, Jul, Oct (within 45 days of quarter end, 60 for Q4)	Gaps of 5–10% are common in large caps. PLAN already blocks entries near results
F&O expiry	NSE: **Tuesday**. Monthly stock and index contracts expire on the last Tuesday (since Sep 2025)	Volatility and pinning near expiry in F&O stocks. Add an `expiry_day` feature
Index rebalancing	Nifty: semi-annual (Mar/Sep); MSCI and FTSE reviews	Flows into stocks being added and out of those being removed, on the effective day
Corporate actions	Ex-dates	Splits, bonuses and dividends create fake gaps in unadjusted data (§9)

8b. The opening: why it matters and what to record (owner, 23 Oct 2026)

In India most of a large cap's daily range is often decided around the open. Overnight news, US and Asian moves, GIFT Nifty, crude, the rupee and last evening's FII/DII flows all land in one place: the **09:00–09:15 pre-open auction**, then the first 15–30 minutes of trading. The owner's rule: track all of it, every day.

WINDOW (IST)	WHAT HAPPENS	WHAT TO RECORD
Previous evening 18:00	FII/DII provisional cash flows published. Results and board-meeting outcomes released after 15:30	FII net, DII net (₹ cr); per-symbol announcements
Overnight to 08:45	US close, US futures, Asia opens (Nikkei 05:30, Kospi 05:30, Hang Seng 06:45 IST), GIFT Nifty trading	S&P 500 / Nasdaq / Dow close %, ES/NQ futures %, **Europe close** (DAX, FTSE 100, CAC 40, Euro Stoxx 50), Nikkei/Hang Seng/Kospi %, USD/INR, DXY, Brent, gold, US 10y, **GIFT Nifty change vs yesterday's Nifty close**, India VIX close, overnight headlines with event tag and sentiment score
During the Indian session	Europe opens about 12:30 IST (summer) / 13:30 IST (winter). US data at 18:00–19:00 IST. US futures trade all day	Europe live %, US futures live %, Brent and USD/INR live, breaking headlines (war, tariffs, central banks) with time stamps so their effect on Nifty can be measured afterwards
09:00–09:08/10	Pre-open order entry, random close	Per stock and for Nifty: indicative equilibrium price (IEP), change vs previous close, total buy and sell quantity (the imbalance), snapshots every 30 s
09:10–09:15	Matching; the open is fixed	Final open price = official open; gap % per stock and for Nifty
09:15–09:45	Opening drive	Opening range (15 and 30 min), first-5-min volume vs its 20-day average, gap filled or not (and when), breadth (advancers vs decliners in the Nifty 100), sector moves
By 15:30	Did the open predict the day?	Open-to-close return, close vs opening range, **opening type** label: gap-up-and-go, gap-up-fade, gap-down-recover, gap-down-go, flat

Major-event news to tag (extends `news/india_rules.py`, which already has rbi_policy, budget, sebi, election, war, fed, inflation, earnings, fii_dii, credit_rating, banking_stress, commodity): `geopolitics_india` (border, Pakistan, China), `middle_east` (Iran, Israel, Red Sea, Hormuz; it moves crude, which India imports), `russia_ukraine`, `tariffs_trade` (US tariffs and trade deals), `us_data` (CPI, payrolls, GDP), `china`, `opec`, `monsoon` (IMD), `gst_council`, `india_macro` (CPI, IIP, GDP), `block_deal`, `promoter_pledge`. Each headline keeps its source, time, tag, symbol hint and score. **Risk-off alarm** (information only): a war/geopolitics headline **and** Brent up 3% or more **and** India VIX up 10% or more within the same 2 hours → red banner on the portal and one Telegram alert.

Two warnings from the crypto engine. First, news sentiment there simply echoed price (the market-news score followed BTC's 24 h move 48 times out of 48). Second, no filter improved the swing book. So the opening record starts as **information and research data**, not a trade filter, until the lab shows lift. The data itself is precious: pre-open snapshots and live breadth can't be downloaded afterwards, so recording starts as early as possible.

Free sources (all unofficial; **VERIFY** each and expect breakage): NSE website JSON for pre-open (`/api/market-data-pre-open?key=NIFTY` and `key=FO`) and FII/DII (`/api/fiidiiTradeReact`), which need a browser-like session cookie. Yahoo chart API for US/Asia indices, futures, USD/INR (`INR=X`), Brent (`BZ=F`), Nifty (`^NSEI`) and India VIX (`^INDIAVIX`). GIFT Nifty has no reliable free API (NSE IX website or a paid feed). Until one is found, US futures plus Asia stand in as the overnight proxy. Angel's WebSocket gives live LTP and quotes from 09:15. Whether it streams indicative prices during the pre-open is **VERIFY**.

9. Data: what to use, and the traps

NEED	SOURCE	TRAPS
Instruments, tokens, tick sizes	Angel scrip master JSON (daily)	Tick sizes are in paise. Ticks change monthly
Daily OHLCV, research history	NSE **bhavcopy**	Since July 2024 the file is the **UDiFF** format (`BhavCopy_NSE_CM_0_0_0_YYYYMMDD_F_0000.csv.zip`, new column names such as `TckrSymb`, `SctySrs`, `OpnPric`, `ClsPric`; Cursor checks the exact headers against a real file). The legacy format applies only to older dates. The parser in `data/history.py` reads only legacy columns

Delivery quantity	NSE "full bhavcopy with delivery" (`sec_bhavdata_full_DDMMYYYY.csv` , **VERIFY** the name)	A separate file from the price bhavcopy
Split / bonus / dividend adjustment	NSE corporate actions, or the yfinance adjusted close (research only)	**Bhavcopy and the Yahoo chart API are unadjusted.** A 1:1 bonus looks like a –50% gap and fires every stop in a backtest. Keep raw prices and an adjustment factor
Index members over time	niftyindices.com constituent lists + NSE index-change notices	Backtesting today's Nifty 100 over 2012–2026 inflates results (survivorship)
1-minute history	Angel `getCandleData`	Users report a 500-candle cap per request and AB1021 throttling. P1-3 measures the real depth
Live ticks, bid/ask	Angel WebSocket V2, SnapQuote mode	Up to 1,000 tokens per connection
Live quotes for paper marking	Angel `ltpData` / full quote	The Yahoo chart API (current `data/free_quotes.py`) is delayed, unofficial and can disappear. It's fine for a ticker tape, not for fills
India VIX, Nifty history	NSE historical index data / Angel index tokens	
FII/DII flows	NSE daily report	Provisional at 18:00, final later
Surveillance lists, holidays, circulars	NSE website	

10. Base rates and strategy evidence

- **SEBI study (FY 2022-23):** 71% of individual intraday traders in the cash segment lost money. Among those making more than 500 trades a year, it was 80%. Under-30s did worst (76%). Costs were a large part of it.
- **The owner's crypto engine:** intraday direction was a coin flip (walk-forward AUC 0.52). The only lasting edge was multi-day breakouts (+0.35R a trade after fees over 5 years). AI judgement and news filters never improved it.
- **Momentum in India:** academic work finds both time-series and cross-sectional momentum in Indian stocks after standard risk adjustments, with sharp crashes in crisis and recovery phases. NSE's Nifty200 Momentum 30 index (6- and 12-month volatility-adjusted momentum, rebalanced twice a year) has led other large-cap indices on rolling returns. It also had the deepest drawdowns, including about –30% from Sep 2024 to Mar 2025.
- **Post-earnings drift:** mixed. Older, longer studies (2002–2011) find drift, especially after bad news. A smaller 2014–2018 study finds none. Treat it as a hypothesis to test, not an edge.
- **Conclusion for the design:** keep the crypto lesson. Daily-bar swing or momentum on liquid large caps is where an edge is plausible and costs are smallest per unit of risk. Intraday stays a shadow experiment until it passes the gate.

11. Glossary

CNC: delivery product. MIS: intraday product. CAS: closing auction session. VaR / ELM: value-at-risk and extreme-loss margins. DP charge: depository fee on a delivery sell. STT: securities transaction tax. Bhavcopy: NSE's daily price file. UDiFF: NSE's unified file format (since 2024). ASM / GSM: surveillance lists. T+1: settlement one trading day after the trade. Tick: the smallest price step. Static IP: the one IP address allowed to send API orders. TOPS: threshold orders per second (10). GIFT Nifty: offshore Nifty futures.

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